

Essential Histology

Question Bank

400 single-best-answer questions with full rationales

Definitive Edition

Companion to **Essential Histology — A Concept-Based Guide for PGME**

Essential Histology: Question Bank

A Concept-Based Guide for PGME — Assessment Companion

Edition: Essential Histology 1405 — Definitive Edition

Primary reference: Mescher AL, Junqueira's Basic Histology: Text and Atlas, 17th Edition, McGraw Hill.

Scope: The 16 official core chapters of the Afghanistan 1405 Medical Specialty Examination syllabus.

How to Use This Bank

This bank contains **417 single-best-answer questions**, written against the learning objectives of the companion teaching text. Every item carries:

- the **correct answer**;
- an explanation of **why the correct option is right**;
- an explanation of **why each remaining option is wrong**;
- a **difficulty level** (Easy / Medium / Hard);
- the **chapter** and **learning objective** it assesses.

Every learning objective stated in the textbook is assessed by at least one item: there are **no orphan objectives**. No question tests a fact that is not taught in the textbook. Distractors reproduce the specific confusions addressed by the textbook's "Do not confuse with" sections, so a wrong answer diagnoses a particular misunderstanding rather than merely recording a lost mark.

Difficulty mix. Easy 104 · Medium 215 · Hard 98. Easy items check that a definition or decisive feature has been learned; Medium items require application of a mechanism; Hard items require integration across topics or discrimination between two closely related entities.

Answer-key balance. Correct answers are distributed evenly across options (A 105 · B 104 · C 104 · D 104), so position gives no clue to the answer.

A note on evolving science. A few items address points where the classical textbook answer has been revised by newer evidence — the embryological origin of thyroid C cells, the status of the 30-nm chromatin fibre, and serous demilunes. These are flagged in their explanations, which state both the current evidence and the older answer that some examinations still expect.

Chapter 1 — Histology and Methods of Study

Q1. In H&E, nuclei stain blue because:

- A) Nucleic acid phosphates are anionic and bind the cationic dye haematoxylin
- B) Nuclear proteins bind haematoxylin
- C) Eosin is a basic dye with affinity for DNA
- D) DNA is cationic and binds eosin

Answer: A · Difficulty: Easy · **LO 1.4** (staining chemistry)

Why A: Basophilia means affinity for a basic (cationic) dye; the phosphate backbone of DNA/RNA is the anion responsible.

Why not: B) Proteins are largely cationic and bind eosin. C) Eosin is acidic/anionic. D) Reverses both charges.

Q2. Which stain requires a frozen section?

- A) Prussian blue B) Oil Red O C) PAS D) Congo red

Answer: B · Difficulty: Easy · **LO 1.4**

Why B: Oil Red O is a lysochrome that partitions into lipid; paraffin processing dissolves lipid in alcohol and xylene, so lipid must be preserved by freezing.

Why not: A, C, D all work on paraffin sections — iron, carbohydrate, and amyloid are not extracted by solvents.

Q3. A liver biopsy shows magenta cytoplasmic granules with PAS that disappear after diastase pre-treatment. The material is:

- A) Mucin B) Basement membrane C) Glycogen D) α 1-antitrypsin polymer

Answer: C · Difficulty: Medium · **LO 1.4**

Why C: Diastase (amylase) digests glycogen specifically, so PAS positivity that is abolished by diastase identifies glycogen.

Why not: A, B, D are all PAS-positive but **diastase-resistant**. α 1-antitrypsin globules are the classic PAS-positive diastase-resistant finding.

Q4. Von Kossa staining is best described as demonstrating:

- A) Calcium directly by chelation
- B) Acidic mucins
- C) Collagen fibres
- D) Phosphate and carbonate anions, with calcium inferred indirectly

Answer: D · Difficulty: Medium · **LO 1.4**

Why D: Silver replaces the cation bound to phosphate/carbonate and is reduced to visible metallic silver. Mineralisation is demonstrated; calcium is inferred.

Why not: A) Alizarin red detects calcium more directly. C) Trichrome/silver reticulin stains target collagen. B) Alcian blue targets acidic mucins.

Q5. The practical resolution limit of the light microscope ($\sim 0.2 \mu\text{m}$) is set principally by:

A) The wavelength of visible light and numerical aperture B) Magnification available C) Lens grinding quality D) Section thickness

Answer: A · Difficulty: Medium · LO 1.5

Why A: The Abbe relationship makes resolution proportional to wavelength and inversely proportional to numerical aperture.

Why not: B, C) Beyond the resolution limit, more magnification gives "empty magnification." D) Thickness affects clarity, not the theoretical limit.

Q6. Microvilli appear as an unresolved "brush border" by light microscopy because:

A) They are not stained by eosin
B) Their $\sim 0.1 \mu\text{m}$ width is below the resolution limit
C) They are destroyed during processing
D) They are only present after fixation

Answer: B · Difficulty: Medium · LO 1.5

Why B: Individual microvilli are narrower than $0.2 \mu\text{m}$ and cannot be separated; they sum into a fuzzy band.

Why not: A) They are eosinophilic. C) They survive processing. D) They are native structures.

Q7. The chief reason immunohistochemistry on formalin-fixed tissue often requires heat-induced epitope retrieval is:

A) Formalin extracts antigens
B) Paraffin binds antibodies
C) Methylene cross-links mask epitopes
D) Formalin lowers tissue pH irreversibly

Answer: C · Difficulty: Medium · LO 1.2

Why C: Formaldehyde forms methylene bridges between proteins, sterically concealing antibody binding sites; heat reverses much of this.

Why not: A) Antigens are retained, not extracted. B) Paraffin is removed before staining. D) pH effects are not the principal mechanism.

Q8. Glutaraldehyde is preferred over formalin for electron microscopy because it:

A) Penetrates tissue faster
B) Preserves lipids without osmium
C) Preserves epitopes better
D) Cross-links more extensively, better preserving ultrastructure

Answer: D · Difficulty: Medium · LO 1.2

Why D: Being a dialdehyde it forms more cross-links and stabilises fine structure.

Why not: A) It penetrates poorly, hence 1 mm blocks. B) Osmium is still needed for lipid/membranes. C) It destroys many epitopes.

Q9. Adipose tissue appears as empty vacuoles in routine sections because:

- A) Lipid is not fixed by formaldehyde and is dissolved in alcohol and xylene
- B) Adipocytes rupture during sectioning
- C) Lipid is eosinophilic but very pale
- D) Adipocytes contain no cytoplasm

Answer: A · Difficulty: Easy · LO 1.2

Why A: Formaldehyde does not fix lipid; processing solvents extract it, leaving the space.

Why not: B) The cell outline survives. C) Lipid is not present to stain. D) B thin cytoplasmic rim remains.

Q10. A 3 cm specimen fixed whole in formalin for 2 hours shows well-preserved periphery but blurred nuclei and detached epithelium centrally. The explanation is:

- A) Over-fixation of the centre
- B) Penetration-limited fixation with central autolysis
- C) Ice crystal artifact
- D) Knife chatter

Answer: B · Difficulty: Medium · LO 1.2

Why B: Formalin penetrates about 1 mm/hour; the centre remains unfixed and autolyses. Large specimens must be sliced.

Why not: A) Over-fixation causes hardening, not autolysis. C, D) Are technical artifacts with different appearances.

Q11. Congo red demonstrating amyloid must be examined under:

- A) Phase contrast B) Dark field C) Polarised light for apple-green birefringence D) UV without filters

Answer: C · Difficulty: Easy · LO 1.4

Why C: The ordered β -pleated sheet aligns the dye to give apple-green birefringence, which is the specific criterion.

Why not: A, B, D do not produce the diagnostic birefringence.

Q12. Which is the correct statement about PAS and basement membranes?

- A) PAS detects type IV collagen directly
- B) PAS positivity of basement membrane is diastase-sensitive
- C) PAS is a specific collagen stain

D) PAS is positive because of carbohydrate-rich glycoproteins and proteoglycans

Answer: D · Difficulty: Hard · LO 1.4

Why D: Laminin, perlecan, nidogen and associated glycoproteins carry the 1,2-glycol groups oxidised by periodic acid.

Why not: A, C) PAS is a carbohydrate stain, not a collagen stain. B) Basement membrane PAS positivity is diastase-resistant.

Q13. Transmission electron microscopy differs from light microscopy fundamentally because:

- A) Electrons have a far shorter wavelength
- B) Its lenses are better ground
- C) It uses higher magnification settings
- D) It requires thicker sections

Answer: A · Difficulty: Easy · LO 1.5

Why A: Wavelength governs resolution; accelerated electrons have wavelengths thousands of times shorter than light.

Why not: B, C) Neither overcomes a wavelength limit. D) TEM uses 60–90 nm ultrathin sections.

Q14. A surgeon requests an intraoperative margin assessment. The appropriate technique is:

- A) Paraffin section with H&E
- B) Frozen section
- C) Electron microscopy
- D) Decalcified section

Answer: B · Difficulty: Easy · LO 1.3

Why B: Frozen section gives an answer in minutes, which is the operative requirement.

Why not: A) Takes many hours to overnight. C) Far too slow and narrow-field. D) Only for mineralised tissue, and slow.

Q15. Which artifact is recognised by the fact that it crosses tissue boundaries without respecting them?

- A) Granuloma B) Necrosis C) Knife chatter lines D) Metaplasia

Answer: C · Difficulty: Easy · LO 1.6

Why C: Technical artifacts ignore biology; regular parallel lines crossing all structures indicate blade vibration.

Why not: A, B, D are biological processes that respect tissue architecture.

Q16. A metachromatic reaction (toluidine blue turning purple on mast cell granules) occurs because:

- A) The granules are eosinophilic
- B) Mast cells contain melanin

- C) The dye is oxidised by heparin
- D) Densely packed polyanions alter the dye's absorption spectrum

Answer: D · Difficulty: Hard · LO 1.4

Why D: High-density sulfated glycosaminoglycan (heparin) causes dye molecules to stack, shifting colour.

Why not: A) Metachromasia is unrelated to eosin. C) No oxidation occurs. B) Mast cells contain no melanin.

Q17. Reticular fibres are demonstrated by silver impregnation because they are:

- A) Argyrophilic — silver is deposited and reduced by an external reducing agent
- B) Argentaffin — they reduce silver themselves
- C) Birefringent
- D) Strongly eosinophilic

Answer: A · Difficulty: Hard · LO 1.4

Why A: Reticular fibre glycoprotein coats bind silver, which requires an added reducer — the definition of argyrophilic.

Why not: B) Melanin and some enterochromaffin granules are argentaffin. C) Not the basis of the stain. D) They are nearly invisible in H&E.

Q18. A tubular gland cut tangentially through its wall may appear as:

- A) A ring with a lumen
- B) A solid mass of tissue without a lumen
- C) Two parallel walls
- D) A single layer of squamous cells

Answer: B · Difficulty: Medium · LO 1.6

Why B: A plane that grazes the wall without entering the lumen shows only wall tissue.

Why not: A) Transverse section. C) Longitudinal section. D) Describes an epithelium, not a plane effect on a gland.

Q19. A cuboidal epithelium may be mistaken for simple squamous when:

- A) The stain is too strong
- B) The tissue is frozen
- C) The section plane passes above or below the level of the nuclei
- D) The magnification is too high

Answer: C · Difficulty: Medium · LO 1.6

Why C: Without nuclei in the plane, the residual cytoplasm looks like a thin flat layer.

Why not: A, B, D do not change apparent cell height in this systematic way.

Q20. The decisive test distinguishing pseudostratified from stratified epithelium is:

A) Nuclear height B) Presence of cilia C) Number of visible nuclear rows D) Whether every cell contacts the basement membrane

Answer: D · Difficulty: Medium · LO 1.6

Why D: Pseudostratification is by definition a single layer with nuclei at different heights, all cells reaching the base.

Why not: A, C) Both are misleading precisely because of nuclear staggering. B) Cilia are common but not definitional.

Q21. The erythrocyte is used as an internal scale reference because its diameter is approximately:

A) 7.5 μm B) 2 μm C) 15 μm D) 25 μm

Answer: A · Difficulty: Easy · LO 1.5

Why A: $\sim 7.5 \mu\text{m}$, conveniently similar to a small lymphocyte nucleus.

Why not: B) Platelet scale. C) Monocyte scale. D) Larger than most leukocytes.

Q22. Osmium tetroxide is used in EM preparation because it:

A) Dissolves lipid B) Fixes and stains unsaturated lipids, rendering membranes electron-dense C) Removes protein D) Decalcifies tissue

Answer: B · Difficulty: Medium · LO 1.2

Why B: It reacts with carbon-carbon double bonds in membrane lipids and adds electron density.

Why not: A) The opposite. C, D) Not its function.

Q23. Masson trichrome is most useful when the clinical question is:

A) Is there iron? B) Is there amyloid? C) How much fibrosis is present? D) Is there glycogen?

Answer: C · Difficulty: Easy · LO 1.3

Why C: Trichrome separates collagen (blue/green) from muscle and cytoplasm (red), quantifying fibrosis.

Why not: A) Perls. B) Congo red. D) PAS with diastase.

Q24. A pathologist must distinguish alcoholic fatty liver from glycogen storage disease on a biopsy where both show pale, swollen hepatocytes. The correct pair of tests is:

A) Congo red and Perls

B) Trichrome and reticulin

C) Von Kossa and Alcian blue

D) Oil Red O on frozen section, and PAS with diastase

Answer: D · Difficulty: Hard · LO 1.3

Why D: Lipid requires a frozen section with a lysochrome; glycogen is identified by diastase-sensitive PAS positivity. Together they discriminate the two.

Why not: A) Amyloid and iron. C) Mineral and acidic mucin. B) Fibrosis and architecture.

Q25. In a systematic approach to an unknown slide, the correct first step is to:

- A) Use scanning power to establish overall architecture
- B) Switch to 40× and identify cells
- C) Order special stains
- D) Count mitoses

Answer: A · Difficulty: Easy · LO 1.6

Why A: Architecture constrains the possibilities before any cell is named; cells confirm rather than initiate the diagnosis.

Why not: B) Premature high power is the commonest student error. C, D) Come much later, if at all.

Q26. Living tissue must be fixed before examination primarily because:

- A) Autolytic enzymes released from lysosomes begin destroying cellular architecture within minutes of death
- B) It is too soft to cut
- C) It lacks colour
- D) It is too thick for light to pass through

Answer: A · Difficulty: Easy · LO 1.1

Why A: Fixation cross-links protein and arrests enzymatic self-digestion, preserving structure as it was in life.

Why not: B) Hardness is addressed by embedding. C) Addressed by staining. D) Addressed by sectioning. Each of these is a later step that would be pointless on already-autolysed tissue.

Q27. A biopsy is processed by paraffin embedding and stained with H&E. Compared with a frozen section of the same tissue, the paraffin section will:

- A) Preserve lipid better
- B) Give superior morphological detail but take hours and lose lipid
- C) Be available within minutes
- D) Preserve enzyme activity

Answer: B · Difficulty: Medium · LO 1.7

Why B: Full processing gives thin, well-preserved sections but requires dehydration and clearing, which extract lipid, and takes hours to overnight.

Why not: A, D) Frozen sections preserve lipid and enzyme activity. C) That is the advantage of the frozen section.

Q28. Under polarised light a Congo red-stained deposit glows apple-green. The correct interpretation is:

- A) Glycogen is present

B) Iron is present

C) Amyloid is present, because its β -pleated sheet aligns the dye

D) Calcium is present

Answer: C · Difficulty: Medium · LO 1.8

Why C: Birefringence arises from the ordered β -pleated sheet; the finding is the diagnostic criterion for amyloid.

Why not: A) PAS, magenta. B) Perls, blue. D) Alizarin red (or von Kossa indirectly).

Chapter 2 — Cytoplasm

Q1. Intensely basophilic cytoplasm indicates abundant:

A) Mitochondria B) Rough ER and free ribosomes C) Smooth ER D) Lipid droplets

Answer: B · Difficulty: Easy · LO 2.1

Why B: Ribosomal RNA is the polyanion binding haematoxylin.

Why not: A) Mitochondria give eosinophilia. C, D) Give pale, foamy cytoplasm.

Q2. A cell with pale, foamy, vacuolated cytoplasm and tubular mitochondrial cristae is most likely secreting:

A) Antibody B) Digestive enzymes C) Steroid hormone D) Mucus

Answer: C · Difficulty: Medium · LO 2.3

Why C: Tubular cristae house steroidogenic P450 enzymes; lipid droplets (extracted in processing) hold cholesterol precursor.

Why not: A, B) Would be strongly basophilic with RER. D) Mucin granules give a different, apically clustered pale appearance.

Q3. The perinuclear "hof" of a plasma cell corresponds to:

A) Nucleolus B) Centriole pair only C) Lysosome cluster D) Golgi apparatus

Answer: D · Difficulty: Easy · LO 2.2

Why D: The Golgi is large in professional secretory cells but unstained in H&E, appearing as a pale zone.

Why not: A) Is intranuclear. C) Would not be uniformly pale. B) Too small to see by light microscopy.

Q4. The signal recognition particle functions to:

A) Pause translation and target the ribosome to the RER translocon

B) Cleave the signal peptide

C) Add mannose-6-phosphate

D) Fold nascent protein

Answer: A · Difficulty: Medium · LO 2.2

Why A: SRP binds the emerging signal sequence, halts elongation, and docks on the SRP receptor.

Why not: B) Signal peptidase does this. C) GlcNAc-phosphotransferase in cis-Golgi. D) Chaperones such as BiP.

Q5. In I-cell disease the fundamental defect is failure to:

A) Synthesise lysosomal enzymes

B) Tag lysosomal enzymes with mannose-6-phosphate

C) Fuse lysosomes with phagosomes

D) Acidify the lysosome

Answer: B · Difficulty: Hard · LO 2.2

Why B: Deficient GlcNAc-phosphotransferase means no M6P sorting signal; enzymes are secreted rather than delivered.

Why not: A) Enzymes are made normally — serum levels are high. C, D) Not the primary lesion.

Q6. The diagnostic biochemical paradox of I-cell disease is:

- A) Low serum and low cellular enzyme
- B) High cellular enzyme with low serum enzyme
- C) High serum enzyme with deficient intracellular enzyme
- D) Normal enzyme with abnormal substrate

Answer: C · Difficulty: Hard · LO 2.2

Why C: Mis-sorted enzymes are secreted into plasma while lysosomes remain enzyme-poor.

Why not: A, B, D misstate the sorting defect.

Q7. In α 1-antitrypsin deficiency, the liver injury results from:

- A) Absence of the protein in hepatocytes
- B) Autoimmune attack on hepatocytes
- C) Excess protease in the liver
- D) Retention of misfolded polymer in hepatocyte RER (toxic gain of function)

Answer: D · Difficulty: Hard · LO 2.2

Why D: PiZ protein polymerises and is retained, producing PAS-positive diastase-resistant globules and hepatocyte injury.

Why not: A) That mechanism (loss of function) explains the lung disease. B, C) Not operative.

Q8. Panacinar emphysema in α 1-antitrypsin deficiency occurs because:

- A) Elastase destroys alveolar elastin unopposed
- B) Collagen is overproduced
- C) Surfactant is deficient
- D) Mucus plugs the airways

Answer: A · Difficulty: Medium · LO 2.2

Why A: Absent circulating antiprotease permits neutrophil elastase to degrade alveolar wall elastin.

Why not: B, C, D describe other lung diseases.

Q9. Tubular mitochondrial cristae are characteristically seen in:

- A) Cardiac myocytes
- B) Adrenal cortical cells
- C) Plasma cells
- D) Neutrophils

Answer: B · Difficulty: Medium · LO 2.3

Why B: Steroid-secreting cells show tubulovesicular cristae.

Why not: A) Lamellar cristae, very numerous. C, D) Not steroidogenic.

Q10. Heteroplasmy explains why mitochondrial diseases:

- A) Are inherited in an autosomal dominant pattern
- B) Affect only males
- C) Show variable severity and threshold effects in high-demand tissues
- D) Always present at birth

Answer: C · Difficulty: Hard · LO 2.3

Why C: A variable proportion of mutant mtDNA per cell means disease appears only above a tissue-specific threshold.

Why not: A) Inheritance is maternal. B) Both sexes affected. D) Onset is variable.

Q11. Intermediate filament proteins are diagnostically valuable in tumour pathology because they:

- A) Are the strongest filaments
- B) Are present only in cancer
- C) Bind antibodies non-specifically
- D) Are lineage-specific and persist despite dedifferentiation

Answer: D · Difficulty: Medium · LO 2.4

Why D: Cytokeratin/vimentin/desmin/GFAP identify epithelial, mesenchymal, muscle, and glial lineage respectively.

Why not: A) Irrelevant to diagnosis. C) The opposite is required. B) They are normal constituents.

Q12. A poorly differentiated metastasis stains positive for cytokeratin. The lineage is:

- A) Epithelial (carcinoma) B) Mesenchymal (sarcoma) C) Glial D) Lymphoid

Answer: A · Difficulty: Easy · LO 2.4

Why A: Cytokeratins are the intermediate filaments of epithelium.

Why not: B) Vimentin. C) GFAP. D) CD45/LCA.

Q13. The axoneme of a motile cilium has the arrangement:

- A) 9+0 B) 9+2 C) 9 triplets D) 13 protofilaments only

Answer: B · Difficulty: Easy · LO 2.4

Why B: Nine outer doublets plus a central pair.

Why not: A) The non-motile primary cilium. C) Centriole/basal body. D) Describes a single microtubule wall.

Q14. Kartagener syndrome results from a defect in:

- A) Kinesin B) Actin C) Dynein arms D) Lamin A

Answer: C · Difficulty: Medium · LO 2.4

Why C: Absent dynein arms prevent doublet sliding and ciliary beating.

Why not: A) Anterograde microtubule motor, not the ciliary bending motor here. B, D) Unrelated systems.

Q15. Situs inversus occurs in primary ciliary dyskinesia because:

- A) Cilia determine organ size

- B) Mucus accumulates in the chest
- C) Sperm cannot swim
- D) Nodal cilia establish embryonic left-right asymmetry

Answer: D · Difficulty: Hard · LO 2.4

Why D: Without directional nodal flow, laterality is randomised, so ~50% of patients have situs inversus.

Why not: A) False. B, C) Are other features of the same syndrome, not the cause of laterality defects.

Q16. The primary cilium differs from a motile cilium in that it is:

- A) 9+0, solitary and sensory B) 9+2 and motile C) Composed of actin D) Found only in the airway

Answer: A · Difficulty: Medium · LO 2.4

Why A: B solitary non-motile sensory antenna present on most cells; dysfunction underlies polycystic kidney disease.

Why not: B) Describes motile cilia. C) Microvilli are actin-cored. D) It is near-ubiquitous.

Q17. Colchicine and vinca alkaloids arrest mitosis by acting on:

- A) Actin B) Microtubules C) Intermediate filaments D) Nuclear lamina

Answer: B · Difficulty: Easy · LO 2.4

Why B: They prevent tubulin polymerisation, so the spindle cannot form.

Why not: A) Cytochalasin/phalloidin. C) No pharmacological agents in routine use. D) Not a drug target here.

Q18. The unfolded protein response is triggered by:

- A) Excess correctly folded protein in the Golgi
- B) Mitochondrial depolarisation
- C) Accumulation of misfolded protein in the ER lumen
- D) DNA damage

Answer: C · Difficulty: Medium · LO 2.2

Why C: Chaperone saturation signals translational attenuation, chaperone upregulation, and ultimately apoptosis.

Why not: A) Not a stress. B, D) Trigger other pathways.

Q19. Peroxisomes are distinguished from lysosomes by containing:

- A) Acid hydrolases B) Proton pumps only C) Mannose-6-phosphate receptors D) Catalase and oxidases for β -oxidation of very-long-chain fatty acids

Answer: D · Difficulty: Medium · LO 2.3

Why D: Peroxisomes generate and then degrade H_2O_2 using catalase and handle VLCFA.

Why not: A, C) Lysosomal features. B) Not distinguishing.

Q20. Zellweger syndrome is a disorder of:

A) Peroxisome biogenesis B) Lysosomal enzyme sorting C) Mitochondrial DNA D) Ciliary dynein

Answer: A · Difficulty: Hard · LO 2.3

Why A: PEX gene mutations prevent peroxisome assembly, causing VLCFB accumulation and severe neurological disease.

Why not: B) I-cell disease. C) Mitochondrial cytopathies. D) Ciliopathies.

Q21. Intensely eosinophilic, granular cytoplasm in the gastric parietal cell reflects:

A) Abundant RER B) Abundant mitochondria supplying ATP for proton pumping C) Stored mucin D) Glycogen

Answer: B · Difficulty: Medium · LO 2.1

Why B: Pumping H^+ against a millionfold gradient is energetically enormous.

Why not: A) Would be basophilic. C, D) Are extracted and appear pale.

Q22. The microvillus core is composed of:

A) Microtubules B) Keratin C) Actin filaments D) Desmin

Answer: C · Difficulty: Easy · LO 2.4

Why C: Bundled actin filaments cross-linked by villin and fimbrin, anchored in the terminal web.

Why not: A) Cilia. B, D) Intermediate filaments with structural, not projection-forming, roles.

Q23. Stereocilia in the epididymis are correctly described as:

A) True motile cilia B) Keratin projections C) Microtubule 9+0 structures D) Long, branching microvilli (actin-based)

Answer: D · Difficulty: Medium · LO 2.4

Why D: Despite the name, they are elongated actin-cored microvilli and are non-motile.

Why not: A, C) Both imply an axoneme, which they lack. B) Incorrect protein.

Q24. An electron micrograph shows a cell with extensive RER, a large Golgi, and abundant secretory granules. The best functional description is:

A) Regulated secretion of protein B) Steroid secretion C) Active ion transport D) Phagocytosis

Answer: A · Difficulty: Medium · LO 2.2

Why A: Storage granules indicate a regulated (stimulus-dependent) protein secretory pathway.

Why not: B) Would show SER and lipid, no granules. C) Would show mitochondria and infoldings. D) Would show lysosomes and phagosomes.

Q25. Lipofuscin is best characterised as:

A) An infectious inclusion

- B) Indigestible residue of lipid peroxidation accumulating in lysosomes with age
- C) Stored glycogen
- D) A viral protein aggregate

Answer: B · Difficulty: Medium · LO 2.5

Why B: The golden-brown "wear-and-tear" pigment of long-lived cells such as neurons, cardiac myocytes, hepatocytes.

Why not: A, D) Not infectious. C) Glycogen is PAS-positive and extracted differently.

Q26. Which stain-target pairing for a cytoplasmic structure is correct?

- A) Oil Red O → glycogen
- B) Congo red → mitochondria
- C) Prussian blue → lipid
- D) PAS → glycogen, with loss of staining after diastase

Answer: D · Difficulty: Medium · LO 2.6

Why D: PAS detects the 1,2-glycol groups of carbohydrate; diastase sensitivity identifies the carbohydrate specifically as glycogen.

Why not: A) Oil Red O detects lipid and needs a frozen section. C) Prussian blue detects iron. B) Congo red detects amyloid.

Chapter 3 — Nucleus

Q1. A large, pale, open nucleus with a prominent nucleolus indicates a cell that is:

A) Quiescent B) Apoptotic C) Transcriptionally and synthetically active D) Terminally keratinised

Answer: C · Difficulty: Easy · LO 3.1

Why C: Euchromatin is decondensed and accessible; a large nucleolus means active ribosome production.

Why not: A) Would be small and dense. B) Pyknotic/fragmented. D) Anucleate.

Q2. Heterochromatin appears dark in H&E because it is:

A) Rich in protein only B) Calcified C) Bound to eosin D) Densely packed, concentrating DNA phosphate per unit volume

Answer: D · Difficulty: Medium · LO 3.1

Why D: Compaction raises local anionic density, so more haematoxylin binds per unit area.

Why not: A) DNA is the key binder. C) It is basophilic. B) Not mineralised.

Q3. The number of Barr bodies in a 47,XXY (Klinefelter) cell is:

A) 1 B) 0 C) 2 D) 3

Answer: A · Difficulty: Medium · LO 3.2

Why A: Barr bodies = number of X chromosomes minus one.

Why not: B) 46,XY or 45,X. C) 47,XXX. D) 48,XXXX.

Q4. X-inactivation in females produces mosaicism because it is:

A) Uniform across all cells
B) Random in each cell but clonally inherited thereafter
C) Determined by paternal origin always
D) Reversed at puberty

Answer: B · Difficulty: Hard · LO 3.2

Why B: Random early choice, faithfully propagated, creates patches of two cell populations.

Why not: A) Would preclude mosaicism. C) Random, not imprinted, in humans generally. D) It is stable.

Q5. The current evidence regarding the 30-nm chromatin fibre is best stated as:

A) It is the proven universal structure of interphase chromatin
B) It does not exist under any condition
C) It forms in vitro and in some specialised chromatins, but is not demonstrated as a regular structure in typical interphase nuclei
D) It is composed of microtubules

Answer: C · Difficulty: Hard · LO 3.1

Why C: Cryo-EM, SAXS and Hi-B favour irregularly folded, interdigitated 10-nm fibres in a disordered state in vivo.

Why not: A) Overstates the evidence. B) Understates it — it is demonstrable in vitro and in starfish sperm/avian erythrocytes. D) Wrong polymer entirely.

Q6. Nuclear import of a protein bearing an NLS requires:

A) Exportin and Ran-GDP B) Vesicular transport C) Passive diffusion only
D) Importin, with cargo release by Ran-GTP in the nucleus

Answer: D · Difficulty: Medium · LO 3.3

Why D: Ran-GTP is high in the nucleus (Ran-GEF), causing importin to release cargo there.

Why not: A) Describes export. C) Only for molecules under ~40 kDa. B) The envelope is not crossed by vesicles.

Q7. The nucleolus is basophilic principally because it contains abundant:

A) rRNA B) DNA only C) Lipid D) Calcium

Answer: A · Difficulty: Easy · LO 3.4

Why A: rRNB transcription and ribosome assembly concentrate RNB there.

Why not: B) DNB contributes, but rRNB dominates. C, D) Not nucleolar constituents.

Q8. Nucleolar organiser regions are located on the:

A) Metacentric chromosomes B) Acrocentric chromosomes 13, 14, 15, 21, 22
C) Sex chromosomes D) Chromosome 1 only

Answer: B · Difficulty: Medium · LO 3.4

Why B: Their short arms carry the tandem rRNA gene repeats.

Why not: A, C, D are incorrect assignments.

Q9. Apoptosis is distinguished from necrosis by:

A) Cell swelling and inflammation
B) Karyolysis
C) Cell shrinkage, membrane integrity, and absence of inflammation
D) Neutrophil infiltration

Answer: C · Difficulty: Easy · LO 3.5

Why C: Apoptosis is an ordered, ATP-dependent, caspase-mediated, non-inflammatory process.

Why not: A, B, D are all features of necrosis.

Q10. The intrinsic (mitochondrial) apoptotic pathway is initiated by:

A) Fas ligand binding Fas
B) Perforin insertion
C) Caspase-8 activation at the membrane

D) Cytochrome c release forming the apoptosome and activating caspase-9

Answer: D · Difficulty: Medium · LO 3.5

Why D: Mitochondrial outer membrane permeabilisation releases cytochrome c, which with Apaf-1 forms the apoptosome.

Why not: A, C) Extrinsic death-receptor pathway. B) Cytotoxic T-cell mechanism.

Q11. Coagulative necrosis is characteristic of:

A) Most solid organ ischaemia B) Brain infarction C) Tuberculosis D) Acute pancreatitis

Answer: A · Difficulty: Medium · LO 3.5

Why A: Protein denaturation preserves tissue outlines while nuclei are lost.

Why not: B) Liquefactive. C) Caseous. D) Fat necrosis.

Q12. Liquefactive necrosis is typical of the brain because:

A) The brain has abundant collagen
B) High lipid content and abundant hydrolytic enzymes with little supporting stroma
C) It is avascular
D) It calcifies rapidly

Answer: B · Difficulty: Hard · LO 3.5

Why B: Enzymatic digestion dominates and there is no collagenous framework to preserve outlines.

Why not: A) The opposite. C) It is highly vascular. D) Not a feature.

Q13. Pyknosis refers to:

A) Nuclear fragmentation B) Nuclear dissolution C) Nuclear shrinkage and chromatin condensation D) Nucleolar enlargement

Answer: C · Difficulty: Easy · LO 3.5

Why C: Pyknosis → karyorrhexis (fragmentation) → karyolysis (dissolution) is the sequence.

Why not: A) Karyorrhexis. B) Karyolysis. D) Unrelated.

Q14. The restriction point of the cell cycle occurs in:

A) Early G1 B) G2 C) S phase D) Late G1

Answer: D · Difficulty: Medium · LO 3.6

Why D: After it, the cell is committed to divide irrespective of extracellular signals.

Why not: A) Still signal-dependent. B, C) Already committed.

Q15. Rb protein normally restrains the cycle by:

A) Sequestering E2F transcription factors until phosphorylated B) Degrading cyclins C) Activating CDK1 D) Inducing apoptosis

Answer: A · Difficulty: Hard · LO 3.6

Why A: Cyclin D-CDK4/6 phosphorylates Rb, releasing E2F to drive S-phase genes.

Why not: B) Not its mechanism. C) Rb inhibits progression. D) That is p53's role.

Q16. Cyclin B-CDK1 drives the transition:

A) G1/S B) G2/M C) S/G2 D) M/G1

Answer: B · Difficulty: Medium · **LO 3.6**

Why B: Classically maturation-promoting factor.

Why not: A) Cyclin E-CDK2. C) Cyclin A. D) Governed by cyclin destruction.

Q17. Autophagy is best characterised as:

A) Uncontrolled cell rupture

B) Caspase-driven fragmentation

C) Regulated self-digestion via autophagosome-lysosome fusion

D) Phagocytosis of neighbouring cells

Answer: C · Difficulty: Medium · **LO 3.5**

Why C: Phagophore → autophagosome → autolysosome, with LC3 as the marker; provides substrates and quality control.

Why not: A) Necrosis. B) Apoptosis. D) Heterophagy.

Q18. Nuclear lamins are:

A) Microtubules B) Integral membrane pumps C) Actin filaments D) Intermediate filaments lining the inner nuclear membrane

Answer: D · Difficulty: Medium · **LO 3.1**

Why D: Lamins A, D, C form the lamina, supporting the envelope and tethering heterochromatin.

Why not: A, C) Cytoplasmic systems. B) Not transporters.

Q19. Laminopathies such as Hutchinson-Gilford progeria arise from defects in:

A) LMNA B) COL1A1 C) FBN1 D) DMD

Answer: A · Difficulty: Hard · **LO 3.1**

Why A: Abnormal lamin A (progerin) distorts the nuclear envelope.

Why not: B) Osteogenesis imperfecta. C) Marfan. D) Duchenne muscular dystrophy.

Q20. A "starry sky" pattern in a lymphoid proliferation is produced by:

A) Necrotic debris B) Tingible-body macrophages engulfing apoptotic cells C) Fibrosis D) Amyloid

Answer: B · Difficulty: Medium · **LO 3.5**

Why B: Pale macrophages against a dark background of proliferating cells, reflecting high apoptotic turnover.

Why not: A) Necrosis produces confluent, not scattered pale cells. C, D) Different appearances.

Q21. In malignancy, the increased nuclear-to-cytoplasmic ratio reflects:

- A) Cytoplasmic hypertrophy
- B) Nuclear shrinkage
- C) Nuclear enlargement with relatively scant cytoplasm in rapidly dividing, poorly differentiated cells
- D) Loss of the nucleolus

Answer: C · Difficulty: Medium · LO 3.7

Why C: Proliferating undifferentiated cells invest in nucleus over specialised cytoplasm.

Why not: A, B) Opposite. D) Nucleoli are typically prominent.

Q22. Prominent nucleoli alone are NOT diagnostic of malignancy because they also occur in:

- A) Apoptotic cells B) Erythrocytes C) Keratinocytes of the corneum D) Plasma cells, neurons, hepatocytes and regenerating tissue

Answer: D · Difficulty: Medium · LO 3.7

Why D: Nucleolar prominence means ribosome production, common to any synthetically active cell.

Why not: A) Nucleoli are lost. B, C) Anucleate or nearly so.

Q23. Facultative heterochromatin differs from constitutive heterochromatin in that it:

- A) Can be reactivated in other cell types B) Is never transcribed C) Is found only at centromeres D) Lacks histones

Answer: A · Difficulty: Medium · LO 3.1

Why A: It represents tissue-specific silencing of genes that are active elsewhere.

Why not: B, C) Describe constitutive heterochromatin. D) All chromatin has histones.

Q24. Nuclear pore complexes permit free diffusion of molecules up to approximately:

- A) 4 kDa B) 40 kDa C) 400 kDa D) No passive diffusion occurs

Answer: B · Difficulty: Medium · LO 3.3

Why B: Larger cargo requires signal-mediated, Ran-dependent transport.

Why not: A) Too restrictive. C) Too permissive. D) Small molecules do diffuse.

Q25. Distinguishing apoptosis from necrosis on a section is best achieved by noting that apoptosis affects:

- A) Confluent sheets of cells with neutrophils
- B) Only the nucleus

C) Scattered single cells with intact membranes and no inflammatory infiltrate

D) Only epithelium

Answer: C · Difficulty: Medium · LO 3.5

Why C: Distribution is the decisive feature: single-cell death in an otherwise healthy field.

Why not: A) Describes necrosis. B, D) Neither is true.

Chapter 4 — Epithelial Tissue

Q1. The decisive criterion distinguishing pseudostratified from stratified epithelium is:

- A) Number of nuclear rows B) Presence of cilia C) Cell shape at the surface
- D) Every cell contacts the basement membrane

Answer: D · Difficulty: Easy · LO 4.2

Why D: Pseudostratification is a single layer with staggered nuclei.

Why not: A) Staggered nuclei create the illusion. B) Common but not definitional. C) Used for classifying true stratified epithelia.

Q2. Epithelia are classified by the shape of:

- A) Surface cells B) Basal cells C) The largest cells D) The nuclei

Answer: A · Difficulty: Easy · LO 4.2

Why A: Stratified squamous has cuboidal basal cells; the surface layer names it.

Why not: B, C, D would misclassify most stratified epithelia.

Q3. Urothelium (transitional epithelium) is identified by:

- A) Keratin B) Large dome-shaped, often binucleate umbrella cells C) Goblet cells D) Cilia

Answer: B · Difficulty: Medium · LO 4.3

Why B: Umbrella cells with specialised plaques allow distension and provide an osmotic barrier.

Why not: A) Urothelium is non-keratinised. C, D) Belong to respiratory/intestinal epithelia.

Q4. In pemphigus vulgaris the autoantibody targets desmoglein 3, producing:

- A) A subepidermal blister B) Dermal fibrosis C) An intraepidermal blister with acantholysis D) Basement membrane thickening

Answer: C · Difficulty: Medium · LO 4.4

Why C: Desmosomes join keratinocytes to each other, so the split is within the epidermis and blisters are flaccid.

Why not: A) That is bullous pemphigoid (hemidesmosome). B, D) Not blistering mechanisms.

Q5. Bullous pemphigoid produces a tense, subepidermal blister because the antigens are:

- A) Desmoglein 1 and 3 B) E-cadherin C) Type IV collagen D) BP180 and BP230 of the hemidesmosome

Answer: D · Difficulty: Medium · LO 4.4

Why D: Hemidesmosomes anchor basal cells to the basement membrane, so the whole epidermis lifts off.

Why not: A) Pemphigus. C) Alport/Goodpasture. B) Adherens junction.

Q6. Loss of E-cadherin explains the discohesive, single-file growth of:

A) Lobular breast and diffuse gastric carcinoma B) Squamous cell carcinoma C) Basal cell carcinoma D) Melanoma

Answer: A · Difficulty: Hard · LO 4.4

Why A: CDH1 loss removes adherens-junction cohesion, so cells infiltrate singly rather than as nests.

Why not: B, C, D retain different adhesion profiles and growth patterns.

Q7. Familial hypomagnesaemia with hypercalciuria and nephrocalcinosis results from mutations in:

A) CLDN14 B) CLDN16 or CLDN19 C) Occludin D) ZO-1

Answer: B · Difficulty: Hard · LO 4.4

Why B: Claudin-16/19 mediate paracellular Mg^{2+} and Ca^{2+} reabsorption in the thick ascending limb; CLDN19 disease adds ocular involvement.

Why not: A) CLDN14 is associated chiefly with hearing loss. C, D) Not the causative genes in FHHNC.

Q8. Tight junctions perform a "fence" function, meaning they:

A) Seal the paracellular space only
B) Anchor keratin filaments
C) Prevent mixing of apical and basolateral membrane proteins
D) Permit ionic coupling

Answer: C · Difficulty: Medium · LO 4.4

Why C: Restricting lateral diffusion within the membrane maintains polarity, enabling vectorial transport.

Why not: A) That is the separate "gate" function. B) Desmosomes/hemidesmosomes. D) Gap junctions.

Q9. Gap junction channels are formed by:

A) Claudins B) Integrins C) Cadherins D) Connexins (six forming a connexon)

Answer: D · Difficulty: Easy · LO 4.4

Why D: Two apposed connexons form an aqueous channel for ions and small molecules.

Why not: A) Tight junctions. C) Adherens/desmosomes. B) Cell-matrix adhesion.

Q10. Connexin 26 mutations commonly cause:

A) Autosomal recessive deafness B) Blistering C) Nephrotic syndrome D) Emphysema

Answer: A · Difficulty: Medium · LO 4.4

Why A: Cx26 is required for cochlear potassium recycling.

Why not: B) Junctional adhesion disorders. C) GBM disease. D) Protease imbalance.

Q11. The lamina densa of the basal lamina is composed principally of:

A) Type I collagen B) Type IV collagen network with laminin, nidogen and perlecan C) Elastin D) Type VII collagen

Answer: B · Difficulty: Medium · LO 4.5

Why B: Type IV forms a sheet-like (not fibrillar) network, cross-linked with the laminin network.

Why not: A) Interstitial matrix. C) Elastic fibres. D) Anchoring fibrils below the lamina densa.

Q12. Goodpasture syndrome involves autoantibodies against:

A) Laminin-332
B) Desmoglein 3
C) The NC1 domain of the $\alpha 3$ chain of type IV collagen
D) Acetylcholine receptor

Answer: C · Difficulty: Hard · LO 4.5

Why C: The same $\alpha 3$ chain is present in glomerular and alveolar basement membranes, explaining the renal-pulmonary syndrome.

Why not: B) Pemphigus. A) Junctional EB. D) Myasthenia gravis.

Q13. Alport syndrome is most often inherited as:

A) Autosomal dominant B) Autosomal recessive only C) Mitochondrial D) X-linked (COL4A5)

Answer: D · Difficulty: Medium · LO 4.5

Why D: COL4A5 mutations account for the majority; EM shows basket-weave splitting of the lamina densa.

Why not: A, B) Less common forms exist. C) Not mitochondrial.

Q14. The distinction between carcinoma in situ and invasive carcinoma rests on:

A) Breach of the basement membrane B) Nuclear grade C) Tumour size D) Mitotic count

Answer: A · Difficulty: Easy · LO 4.5

Why A: Until the basement membrane is crossed, the tumour cannot access lymphatics or vessels and cannot metastasise.

Why not: B, C, D influence grade/stage but do not define invasion.

Q15. A brush border is most characteristic of:

A) Respiratory epithelium B) Small intestinal absorptive cells and proximal renal tubule C) Epidermis D) Mesothelium

Answer: B · Difficulty: Easy · LO 4.6

Why B: Dense microvilli greatly amplify absorptive surface area.

Why not: A) Cilia and goblet cells. C) Keratin. D) Simple squamous.

Q16. Serous secretory cells are recognised by:

A) Pale foamy cytoplasm with a flattened basal nucleus

- B) Empty vacuoles
- C) Basophilic base, round nucleus, apical eosinophilic zymogen granules
- D) Keratohyalin granules

Answer: C · Difficulty: Medium · LO 4.7

Why C: High RER for enzyme synthesis, with stored zymogen apically.

Why not: A) Mucous cells. B) Adipocytes. D) Epidermis.

Q17. Holocrine secretion is exemplified by:

- A) Pancreatic acinar cells
- B) Goblet cells
- C) Lactating mammary gland lipid
- D) Sebaceous glands

Answer: D · Difficulty: Easy · LO 4.7

Why D: The entire cell disintegrates and becomes the secretion.

Why not: A, B) Merocrine. C) Apocrine.

Q18. Serous demilunes are now regarded as:

- A) Substantially a fixation artifact of mucous cell swelling, though the term remains in use
- B) A purely native structure in all species
- C) A form of metaplasia
- D) Degenerating acini

Answer: A · Difficulty: Hard · LO 4.7

Why A: Cryofixation studies show intermingled serous and mucous cells without crescents, although the finding remains debated and the term is retained descriptively.

Why not: B) Overstates. C, D) Not pathological processes.

Q19. Epithelium is avascular, which means nutrients must:

- A) Be synthesised locally
- B) Diffuse from capillaries in the underlying connective tissue
- C) Arrive via lymphatics within the epithelium
- D) Come from the lumen only

Answer: B · Difficulty: Easy · LO 4.1

Why B: This diffusion dependence limits epithelial thickness.

Why not: A, C) Epithelia contain no vessels or lymphatics. D) Not a general mechanism.

Q20. Simple squamous epithelium lining body cavities (pleura, peritoneum) is termed:

- A) Endothelium
- B) Urothelium
- C) Mesothelium
- D) Endometrium

Answer: C · Difficulty: Easy · LO 4.3

Why C: Endothelium lines vessels; mesothelium lines serous cavities.

Why not: A) Vascular. B) Urinary tract. D) A whole mucosa, not an epithelial class.

Q21. Anchoring fibrils that secure the lamina densa to underlying connective tissue are composed of:

A) Type I collagen B) Fibrillin C) Laminin D) Type VII collagen

Answer: D · Difficulty: Medium · LO 4.5

Why D: Their defect causes dystrophic epidermolysis bullosa.

Why not: A) Interstitial. C) Within the lamina lucida/densa. B) Elastic system.

Q22. The type of epithelium in the trachea is:

A) Pseudostratified ciliated columnar with goblet cells B) Simple columnar C) Stratified squamous D) Transitional

Answer: A · Difficulty: Easy · LO 4.3

Why A: Adapted for mucociliary clearance.

Why not: B) Found in gut. C) Oesophagus. D) Urinary tract.

Q23. In the glomerular basement membrane, charge selectivity is contributed mainly by:

A) Type IV collagen B) Heparan sulfate proteoglycan (perlecan/agrin) C) Nidogen D) Fibronectin

Answer: B · Difficulty: Hard · LO 4.5

Why B: Its dense anionic charge repels negatively charged plasma proteins such as albumin.

Why not: A) Provides the size-selective scaffold. C) A bridging molecule. D) Not the principal charge barrier.

Q24. An epithelium showing nuclei at multiple levels, cilia, and goblet cells, in which careful inspection shows all cells reaching the base, is:

A) Stratified columnar B) Transitional C) Pseudostratified ciliated columnar D) Stratified squamous

Answer: C · Difficulty: Medium · LO 4.2

Why C: The combination is essentially diagnostic of respiratory epithelium.

Why not: A) Rare, found in some large ducts, non-ciliated. B, D) Different surface morphology.

Q25. Which junction couples cardiac myocytes electrically, permitting the myocardium to act as a functional syncytium?

A) Desmosome B) Hemidesmosome C) Tight junction D) Gap junction

Answer: D · Difficulty: Medium · LO 4.4

Why D: Connexin channels in the intercalated disc allow ionic current to spread cell-to-cell.

Why not: A) Provides mechanical adhesion in the same disc. B, C) Not present in that role.

Chapter 5 — Connective Tissue

Q1. Vitamin C is required for collagen synthesis as a cofactor for:

- A) Prolyl and lysyl hydroxylases
- B) Lysyl oxidase
- C) Procollagen peptidase
- D) Galactosyltransferase

Answer: A · Difficulty: Medium · LO 5.2

Why A: Ascorbate keeps the enzyme iron reduced; hydroxyproline is needed to stabilise the triple helix.

Why not: B) Copper-dependent, acts at cross-linking. C) Cleaves propeptides extracellularly. D) Not vitamin C dependent.

Q2. Lysyl oxidase requires which trace element?

- A) Iron
- B) Copper
- C) Zinc
- D) Selenium

Answer: B · Difficulty: Medium · LO 5.2

Why B: Copper deficiency (Menkes) and lathyrism both impair cross-linking, causing connective tissue and vascular fragility.

Why not: A) Needed by hydroxylases. C, D) Not the cofactor here.

Q3. Scurvy produces bleeding gums and poor wound healing because:

- A) Platelets are deficient
- B) Elastin is absent
- C) Collagen triple helix is unstable without hydroxyproline
- D) Fibrin cannot form

Answer: C · Difficulty: Medium · LO 5.2

Why C: Under-hydroxylated collagen is degraded, weakening vessel walls and new matrix.

Why not: A, D) Coagulation is normal. B) Elastin is not the primary lesion.

Q4. Vascular (type IV) Ehlers-Danlos syndrome involves defective:

- A) Type I collagen
- B) Fibrillin
- C) Type II collagen
- D) Type III collagen

Answer: D · Difficulty: Hard · LO 5.2

Why D: Type III is prominent in vessel walls and bowel; its defect causes arterial and intestinal rupture — the lethal subtype.

Why not: A) Osteogenesis imperfecta. C) Cartilage disorders. B) Marfan.

Q5. Osteogenesis imperfecta most commonly results from mutations in:

- A) COL1A1/COL1A2
- B) COL3A1
- C) FBN1
- D) COL4A5

Answer: A · Difficulty: Medium · LO 5.2

Why A: Defective type I collagen causes fragile bones, blue sclerae, and hearing loss.

Why not: B) Vascular EDS. C) Marfan. D) Alport.

Q6. Reticular fibres are argyrophilic and PAS-positive because they are:

- A) Type I collagen in thick bundles
- B) Type III collagen with an abundant glycoprotein/proteoglycan coat
- C) Composed of elastin
- D) Mineralised

Answer: B · Difficulty: Medium · LO 5.3

Why B: The carbohydrate coat binds silver and reacts with PAS.

Why not: A) Type I is not argyrophilic in this way. C) Elastic stains differ. D) Not mineralised.

Q7. Reticular fibres form the supporting stroma of:

A) Tendon B) Aorta C) Lymph node, spleen, bone marrow and liver D) Cornea

Answer: C · Difficulty: Easy · LO 5.3

Why C: A delicate meshwork supports soft cellular parenchyma while permitting cell and fluid movement.

Why not: A, D) Dense type I collagen. B) Elastic laminae.

Q8. Elastic fibres consist of:

A) Keratin
B) Type I collagen only
C) Proteoglycan aggregates
D) Elastin core on a fibrillin-1 microfibril scaffold

Answer: D · Difficulty: Medium · LO 5.3

Why D: Elastin is cross-linked by desmosine/isodesmosine and deposited on fibrillin microfibrils.

Why not: A, B, C are different matrix components.

Q9. In Marfan syndrome, aortic pathology results from FBN1 mutation causing:

A) Defective microfibril scaffold plus excess TGF- β signalling
B) Excess collagen deposition
C) Elastase deficiency
D) Calcification of the media

Answer: A · Difficulty: Hard · LO 5.3

Why A: Fibrillin-1 normally sequesters latent TGF- β ; loss weakens the media mechanically and dysregulates signalling.

Why not: B, C, D are not the Marfan mechanism.

Q10. Glycosaminoglycans resist compression because they:

A) Are stiff protein rods
B) Are polyanionic, attracting cations and osmotically bound water
C) Cross-link collagen
D) Form fibrils

Answer: B · Difficulty: Medium · LO 5.4

Why B: The hydrated gel creates swelling pressure balanced by the collagen network.

Why not: A, D) They are not fibrillar. C) That is lysyl oxidase's role.

Q11. Hyaluronan differs from other glycosaminoglycans in that it is:

- A) Sulfated and protein-linked
- B) Found only in bone
- C) Unsulfated, very large, and not covalently attached to a core protein
- D) A protein

Answer: C · Difficulty: Hard · LO 5.4

Why C: It forms the backbone to which aggrecan monomers attach via link proteins.

Why not: A) True of the others. B) Widely distributed. D) It is a carbohydrate.

Q12. Hurler syndrome is caused by deficiency of:

- A) Glucocerebrosidase B) Iduronate sulfatase C) Hexosaminidase A D) α -L-iduronidase

Answer: D · Difficulty: Hard · LO 5.4

Why D: Autosomal recessive MPS I with corneal clouding; Hunter (iduronate sulfatase) is X-linked without corneal clouding.

Why not: B) Hunter. C) Tay-Sachs. A) Gaucher.

Q13. Mast cell granules are metachromatic because they contain:

- A) Densely sulfated heparin B) Histamine only C) Melanin D) Lipid

Answer: A · Difficulty: Medium · LO 5.5

Why A: Densely packed polyanions shift toluidine blue from blue to purple.

Why not: B) Histamine is not the metachromatic agent. C, D) Not present.

Q14. Mast cell degranulation in immediate hypersensitivity is triggered by:

- A) Complement only B) Cross-linking of IgE bound to Fc ϵ RI by antigen C) IgG binding D) Direct T-cell contact

Answer: B · Difficulty: Medium · LO 5.5

Why B: Bridging adjacent IgE molecules initiates the signalling that releases preformed and newly synthesised mediators.

Why not: A, C, D are not the classical type I mechanism.

Q15. The plasma cell is recognised by:

- A) Kidney-shaped nucleus B) Bilobed nucleus with red granules C) Eccentric clock-face nucleus with a perinuclear hof D) Multilobed nucleus

Answer: C · Difficulty: Easy · LO 5.5

Why C: Peripheral heterochromatin clumping plus an unstained Golgi.

Why not: A) Monocyte. B) Eosinophil. D) Neutrophil.

Q16. Dense regular connective tissue is distinguished from smooth muscle on H&E by:

- A) Colour of the cytoplasm
- B) Cell size

C) Presence of capillaries

D) Nuclei lying between acellular pink bundles rather than within eosinophilic cells

Answer: D · Difficulty: Medium · LO 5.6

Why D: In tendon, flattened fibroblast nuclei sit in rows between collagen; in muscle, cigar-shaped nuclei lie inside the cells.

Why not: A) Both are pink on H&E — hence the confusion; trichrome separates them. B, C) Unreliable.

Q17. Masson trichrome stains collagen ____ and muscle ____:

A) Blue/green; red B) Red; blue C) Black; yellow D) Yellow; blue

Answer: A · Difficulty: Easy · LO 5.3

Why A: This is the basis for assessing fibrosis.

Why not: B) Reversed. C, D) Wrong stain conventions.

Q18. Dense irregular connective tissue is found in the dermis because it must resist:

A) Tension in one direction only B) Stress from multiple directions C) Compression only D) No mechanical load

Answer: B · Difficulty: Easy · LO 5.6

Why B: Interwoven bundles in many orientations provide multidirectional strength.

Why not: A) Dense regular (tendon). C) Cartilage. D) False.

Q19. The fibroblast is distinguished from the fibrocyte by having:

A) A small, dark, condensed nucleus

B) Phagocytic vacuoles

C) A large, pale, euchromatic nucleus with basophilic cytoplasm

D) Metachromatic granules

Answer: C · Difficulty: Easy · LO 5.5

Why C: The active synthetic form has open chromatin and abundant RER.

Why not: A) The quiescent fibrocyte. B) Macrophage. D) Mast cell.

Q20. Macrophages in connective tissue derive from:

A) Fibroblasts B) Endothelium C) Mast cells D) Blood monocytes

Answer: D · Difficulty: Easy · LO 5.5

Why D: Circulating monocytes enter tissue and differentiate (with an additional embryonic-derived resident population in some organs).

Why not: A, B, C are separate lineages.

Q21. Type II collagen is the principal fibrillar collagen of:

A) Hyaline cartilage B) Bone C) Basement membrane D) Skin

Answer: A · Difficulty: Easy · LO 5.2

Why A: Also present in the vitreous and intervertebral disc nucleus pulposus.

Why not: B, D) Type I. C) Type IV.

Q22. The 67 nm banding of collagen fibrils on EM results from:

- A) Mineral deposits B) Quarter-stagger overlap of tropocollagen molecules
- C) Elastin cross-links D) Proteoglycan spacing

Answer: B · Difficulty: Hard · LO 5.2

Why B: Regular offsetting creates alternating overlap and gap zones that bind stain differently.

Why not: A, C, D do not generate this periodicity.

Q23. Propeptide cleavage of procollagen occurs:

- A) In the RER B) In the Golgi C) Extracellularly D) In the nucleus

Answer: C · Difficulty: Medium · LO 5.2

Why C: Keeping assembly extracellular prevents the cell from being destroyed by intracellular fibril formation.

Why not: A, B) Would cause intracellular assembly. D) Not a site of protein processing.

Q24. Loose (areolar) connective tissue is characterised by:

- A) Few cells and thick parallel bundles
- B) Exclusively adipocytes
- C) Absence of ground substance
- D) Many cell types, sparse randomly arranged fibres, abundant ground substance and capillaries

Answer: D · Difficulty: Easy · LO 5.6

Why D: It lies beneath epithelia and is the main site of the inflammatory response.

Why not: A) Dense regular. B, C) Incorrect.

Q25. A patient with hyperextensible skin, hypermobile joints, and poor wound healing, whose defect lies in removal of procollagen N-propeptides, has:

- A) A form of Ehlers-Danlos syndrome B) Marfan syndrome C) Scurvy D) Osteogenesis imperfecta

Answer: A · Difficulty: Hard · LO 5.2

Why A: Defective propeptide cleavage impairs normal fibril assembly — a recognised EDS mechanism.

Why not: B) FBN1. C) Acquired vitamin deficiency with different features. D) Bone fragility predominates.

Q26. The three structural components of connective tissue are:

- A) Cells, fibres, and ground substance
- B) Collagen, elastin, and reticulin
- C) Epithelium, stroma, and vessels
- D) Cells, vessels, and nerves

Answer: A · Difficulty: Easy · LO 5.1

Why A: The defining feature of connective tissue is that the matrix (fibres plus ground substance), not the cells, is the functional element.

Why not: B) These are three fibre types, only one component. C) Describes organ architecture. D) Omits the matrix entirely.

Q27. A section shows chondrocytes in lacunae within a homogeneous basophilic matrix containing no blood vessels, covered by a perichondrium. This is:

A) Bone B) Hyaline cartilage C) Dense regular connective tissue D) Fibrocartilage

Answer: B · Difficulty: Medium · LO 5.7

Why B: Avascularity, lacunae, a glassy basophilic matrix, and a perichondrium together identify hyaline cartilage.

Why not: A) Bone is vascular with lamellae and canaliculi. C) Tendon has no lacunae. D) Fibrocartilage has visible coarse type I collagen bundles and no perichondrium.

Q28. Fibrocartilage is distinguished from hyaline cartilage by:

A) Having a perichondrium
B) Being more basophilic
C) Containing type I collagen in visible bundles and lacking a perichondrium
D) Being avascular

Answer: C · Difficulty: Hard · LO 5.7

Why C: It is the transitional tissue at tendon-bone and ligament-bone junctions, and the only cartilage built on type I collagen.

Why not: A) It lacks one. B) It is more eosinophilic because of the collagen. D) Both are avascular, so this does not discriminate.

Q29. PTH increases bone resorption by:

A) Binding receptors directly on osteoclasts
B) Stimulating osteocyte division
C) Inhibiting osteoprotegerin synthesis only
D) Acting on osteoblasts, which express RANKL to activate osteoclast precursors

Answer: D · Difficulty: Hard · LO 5.6

Why D: Osteoclasts have **no PTH receptors**. Resorption is commanded by the bone-forming cell — the reason anti-RANKL therapy (denosumab) suppresses resorption.

Why not: A) The classic trap. C) Osteoprotegerin is a RANKL decoy, but this is not the principal mechanism. B) Osteocytes do not divide.

Chapter 6 — Adipose Tissue

Q1. The unilocular white adipocyte is recognised by:

- A) Multiple small vacuoles and a central nucleus
- B) A single large vacuole with a flattened peripheral nucleus
- C) Eosinophilic granular cytoplasm
- D) A clock-face nucleus

Answer: B · Difficulty: Easy · LO 6.1

Why B: The signet-ring profile results from lipid extraction during processing.

Why not: A) Brown fat. C) Mitochondria-rich cells. D) Plasma cell.

Q2. Brown adipose tissue generates heat through:

- A) ATP synthase B) Shivering C) UCP1 (thermogenin) uncoupling oxidation from phosphorylation D) Glycolysis

Answer: C · Difficulty: Medium · LO 6.3

Why C: UCP1 allows protons to bypass ATP synthase, dissipating the gradient as heat.

Why not: A) Captures energy as ATP. B) Muscle mechanism. D) Not the thermogenic pathway here.

Q3. Brown fat is brown because of:

- A) Melanin B) Lipofuscin C) Haemosiderin D) Abundant mitochondrial cytochromes and rich vascularity

Answer: D · Difficulty: Medium · LO 6.3

Why D: Dense mitochondria plus capillaries give the colour.

Why not: A, B, C are other pigments not responsible.

Q4. Leptin acts principally on the:

- A) Hypothalamic arcuate nucleus to suppress appetite B) Liver C) Pancreas D) Adrenal cortex

Answer: A · Difficulty: Easy · LO 6.4

Why A: It is the afferent signal reporting fat mass to the brain.

Why not: B, C, D are not its primary satiety target.

Q5. Common human obesity is characterised by:

- A) Leptin deficiency B) High leptin with leptin resistance C) Absent leptin receptors universally D) Low adipocyte number

Answer: B · Difficulty: Medium · LO 6.4

Why B: This is why leptin therapy is ineffective in ordinary obesity but curative in congenital deficiency.

Why not: A) Rare monogenic form. C) Not universal. D) Adipocyte number is increased or normal.

Q6. Which adipokine falls as adiposity increases and improves insulin sensitivity?

A) Leptin B) TNF- α C) Adiponectin D) Resistin

Answer: C · Difficulty: Medium · LO 6.4

Why C: Low adiponectin is a marker of metabolic risk — the inverse relationship is the distinguishing feature.

Why not: A, B, D all rise with adiposity.

Q7. Crown-like structures in adipose tissue represent:

- A) Multinucleate giant cells of granuloma
- B) Capillary proliferation
- C) Lipoblasts
- D) Rings of macrophages surrounding a dying adipocyte

Answer: D · Difficulty: Hard · LO 6.4

Why D: A direct histological demonstration that obesity is a chronic inflammatory state.

Why not: A) Different context and morphology. C) Liposarcoma. B) Not annular.

Q8. A vacuolated cell whose nucleus is indented and hyperchromatic, within a soft tissue tumour, is:

- A) A lipoblast, indicating liposarcoma B) A normal adipocyte C) A macrophage D) A signet-ring carcinoma cell

Answer: A · Difficulty: Hard · LO 6.5

Why A: Scalloping/indentation of an atypical nucleus by vacuoles is the decisive malignant feature.

Why not: B) Smoothly compressed bland nucleus. C) No large lipid vacuole. D) Vacuole is mucin, PAS/mucicarmine positive.

Q9. Signet-ring cell carcinoma is distinguished from adipocytes by:

- A) Nuclear position B) Intracytoplasmic mucin positive with PAS/mucicarmine and nuclear atypia C) Cell size D) Vacuole number

Answer: B · Difficulty: Medium · LO 6.5

Why B: The content of the vacuole and nuclear features settle it.

Why not: A) Both may be peripheral. C, D) Unreliable.

Q10. Lipodystrophy causes severe insulin resistance because:

- A) There is too much adipose tissue
- B) Leptin is overproduced
- C) Absent adipose storage capacity forces ectopic lipid deposition in liver and muscle
- D) Adipocytes are inflamed

Answer: C · Difficulty: Hard · LO 6.4

Why C: It demonstrates that failure of appropriate fat storage, not fat itself, drives metabolic disease.

Why not: A) The opposite. B) Leptin is low. D) Not the main mechanism.

Q11. Brown adipose tissue in the newborn is concentrated:

- A) In the abdomen only
- B) Subcutaneously and uniformly
- C) In the limbs
- D) Interscapular, perirenal, and around the great vessels

Answer: D · Difficulty: Medium · LO 6.3

Why D: Sited to warm blood returning to the core.

Why not: A, B, C are incorrect distributions.

Q12. Beige (brite) adipocytes:

- A) Appear within white depots in response to cold or β -adrenergic stimulation
- B) Never express UCP1
- C) Are found only in infants
- D) Are identical to white adipocytes

Answer: A · Difficulty: Hard · LO 6.3

Why A: "Browning" is a recognised, potentially therapeutic adaptive response.

Why not: B) They do. C) Present in adults. D) Functionally distinct.

Q13. The statement "brown fat disappears completely after infancy" is:

- A) Correct
- B) Incorrect — metabolically active depots are detectable in adults by PET-CT
- C) Correct only in males
- D) Correct only in cold climates

Answer: B · Difficulty: Medium · LO 6.3

Why B: Supraclavicular and paravertebral depots persist.

Why not: A, C, D perpetuate the outdated claim.

Q14. The sympathetic receptor mediating brown fat thermogenesis is:

- A) $\alpha 1$ B) M3 C) $\beta 3$ D) Nicotinic

Answer: C · Difficulty: Medium · LO 6.3

Why C: Noradrenaline at $\beta 3$ receptors drives lipolysis and UCP1 activity.

Why not: A, B, D mediate other responses.

Q15. Adipose tissue is supported by a delicate network of:

- A) Type I collagen bundles B) Keratin C) Elastic laminae D) Reticular (type III) fibres

Answer: D · Difficulty: Medium · LO 6.1

Why D: Fine reticular fibres surround each adipocyte with abundant capillaries.

Why not: A) Too coarse. B, C) Not the supporting stroma here.

Q16. In H&E, the thin rim of adipocyte cytoplasm appears as:

A) A barely visible eosinophilic line around a clear space B) A thick basophilic band C) Granular brown material D) Nothing at all

Answer: A · Difficulty: Easy · LO 6.1

Why A: Almost the entire cell volume is occupied by extracted lipid.

Why not: B, C) Incorrect appearance. D) B rim is discernible.

Q17. Lipomas are:

A) Malignant B) The commonest benign soft-tissue tumour, composed of mature adipocytes C) Composed of brown fat D) Always multiple

Answer: B · Difficulty: Easy · LO 6.5

Why B: Encapsulated, histologically near-normal fat.

Why not: A) Benign. C) Hibernoma is the brown-fat tumour. D) Usually solitary.

Q18. Sebaceous gland cells may be confused with adipocytes, but are distinguished by:

A) Peripheral nucleus B) Larger size C) Multivacuolated cytoplasm, lobular arrangement around a follicle, and holocrine disintegration D) Lack of vacuoles

Answer: C · Difficulty: Medium · LO 6.2

Why C: Architecture and the dying cells toward the duct are decisive.

Why not: A) Adipocyte feature. B, D) Not discriminating.

Q19. White adipose tissue contributes to thermal insulation and mechanical cushioning, and additionally functions as:

A) A purely inert store B) A lymphoid organ C) A haemopoietic organ in adults D) The largest endocrine organ by mass

Answer: D · Difficulty: Easy · LO 6.4

Why D: Adipokine secretion makes it an endocrine tissue.

Why not: A) Obsolete view. C) Yellow marrow is fat but not haemopoietic. B) Incorrect.

Q20. TNF- α and IL-6 from adipose tissue contribute to insulin resistance by:

A) Impairing insulin receptor signalling
B) Destroying beta cells
C) Blocking glucose absorption
D) Increasing leptin clearance

Answer: A · Difficulty: Medium · LO 6.4

Why A: Chronic low-grade inflammation interferes with post-receptor signalling.

Why not: B) Type 1 mechanism. C, D) Not the pathway.

Q21. Adipocyte mitochondria in brown fat differ from white fat by being:

A) Fewer and smaller B) Numerous, large, with densely packed cristae C) Absent D) Tubular-cristae type

Answer: B · Difficulty: Easy · LO 6.3

Why B: The thermogenic machinery requires abundant mitochondria.

Why not: A) Describes white fat. C) False. D) Steroidogenic cells.

Q22. Fat necrosis following trauma or pancreatitis is characterised histologically by:

A) Caseous debris B) Fibrinoid change C) Saponification — chalky calcium soaps with ghost adipocytes D) Liquefaction only

Answer: C · Difficulty: Hard · LO 6.5

Why C: Released lipase liberates fatty acids that chelate calcium.

Why not: A) TB. B) Vasculitis. D) Brain/abscess.

Q23. The main energy-storage molecule in the adipocyte is:

A) Glycogen B) Phospholipid C) Cholesterol ester D) Triglyceride

Answer: D · Difficulty: Easy · LO 6.1

Why D: Triglyceride yields more energy per gram than carbohydrate and is stored anhydrously.

Why not: A) Liver/muscle store. C) Steroidogenic precursor pool. B) Membrane component.

Q24. Hormone-sensitive lipase in adipocytes is activated by:

A) Catecholamines and glucagon B) Insulin C) Leptin D) Adiponectin

Answer: A · Difficulty: Medium · LO 6.4

Why A: Fasting and stress hormones mobilise stored lipid; insulin is antilipolytic.

Why not: B) Opposes lipolysis. C, D) Not direct activators.

Q25. A section shows lobules of cells with many small vacuoles, round central nuclei, and abundant capillaries, from a neonate's interscapular region. This is:

A) White adipose tissue B) Brown adipose tissue C) Sebaceous gland D) Liposarcoma

Answer: B · Difficulty: Medium · LO 6.2

Why B: Multilocular morphology, central nuclei, site and age are all concordant.

Why not: A) Unilocular with peripheral nuclei. C) Attached to a follicle with a duct. D) Atypical lipoblasts.

Q26. Beige (brite) adipocytes are best described as:

- A) UCP1-capable cells arising within white depots in response to cold or β -adrenergic stimulation
- B) Immature white adipocytes
- C) Brown adipocytes that have lost UCP1
- D) Macrophages containing lipid

Answer: A · Difficulty: Medium · LO 6.6

Why A: "Browning" is an adaptive, inducible response and a current therapeutic target in obesity.

Why not: B) They are a distinct functional state, not an immature stage. C) The reverse. D) Those are foam cells.

Q27. A soft-tissue mass contains cells with multiple cytoplasmic vacuoles that scallop and indent a hyperchromatic, irregular nucleus. The diagnosis is:

- A) Lipoma
- B) Liposarcoma, identified by lipoblasts
- C) Normal brown fat
- D) Hibernoma

Answer: B · Difficulty: Hard · LO 6.7

Why B: The indented atypical nucleus is the decisive malignant feature; a normal adipocyte nucleus is smoothly compressed and bland.

Why not: A) Composed of mature bland adipocytes. C, D) Multivacuolated but with bland central round nuclei and no atypia.

Chapter 7 — Nerve Tissue and Nervous System

Q1. Nissl substance corresponds ultrastructurally to:

A) Mitochondria B) Lysosomes C) Stacked rough ER and free ribosomes D) Neurofilaments

Answer: C · Difficulty: Easy · LO 7.1

Why C: Its basophilia reflects the massive protein synthesis needed to maintain a long axon.

Why not: A) Eosinophilic. B, D) Not basophilic aggregates.

Q2. Nissl substance is absent from:

A) The perikaryon B) Dendrites C) The nucleus D) The axon hillock and axon

Answer: D · Difficulty: Medium · LO 7.1

Why D: The axon has no ribosomes, which is why all axonal protein must be transported from the cell body.

Why not: A, B) Both contain Nissl substance. C) Not a cytoplasmic compartment.

Q3. One oligodendrocyte myelinates:

A) Segments of many different axons B) A single internode of one axon C) No axons D) Only unmyelinated fibres

Answer: A · Difficulty: Medium · LO 7.2

Why A: Processes extend to up to ~50 axons — the key contrast with Schwann cells.

Why not: B) Schwann cell. C, D) Incorrect.

Q4. Peripheral nerves regenerate better than central axons largely because:

A) They are thicker
B) Schwann cells survive and form bands of Büngner guiding regrowth
C) They are unmyelinated
D) They lack connective tissue

Answer: B · Difficulty: Hard · LO 7.2

Why B: The CNS lacks this scaffold and has astrocytic scarring plus myelin-derived inhibitors.

Why not: A, C, D are not the determining factors.

Q5. Microglia differ from all other glia in being derived from:

A) Neuroectoderm B) Neural crest C) Mesoderm/yolk-sac macrophage progenitors D) Endoderm

Answer: C · Difficulty: Medium · LO 7.2

Why C: They are the resident macrophages of the CNS — a classic examination trap.

Why not: A) All other glia. B) Schwann/satellite cells. D) Not a glial source.

Q6. Astrocytes are identified immunohistochemically by:

A) CD68 B) Cytokeratin C) Desmin D) GFAP

Answer: D · Difficulty: Easy · **LO 7.2**

Why D: Glial fibrillary acidic protein is their intermediate filament.

Why not: B) Epithelium. C) Muscle. A) Macrophages/microglia.

Q7. Saltatory conduction depends on concentration of voltage-gated Na⁺ channels at:

A) Nodes of Ranvier B) Internodes C) The soma D) Dendritic spines

Answer: A · Difficulty: Easy · **LO 7.3**

Why A: Depolarisation regenerates only at nodes, so the impulse jumps between them.

Why not: B) Insulated by myelin. C, D) Not the propagation mechanism.

Q8. Myelin appears as a clear halo in routine H&E because:

A) It is basophilic B) Its lipid is dissolved during processing C) It is mineralised D) It shrinks away

Answer: B · Difficulty: Easy · **LO 7.3**

Why B: Luxol fast blue or osmium is required to demonstrate it.

Why not: A) It is not stained. C) Not mineral. D) Not the principal reason.

Q9. The blood-nerve barrier resides in the:

A) Endoneurium B) Epineurium C) Perineurium, via tight junctions between its epithelioid cells D) Schwann cell basal lamina

Answer: C · Difficulty: Hard · **LO 7.4**

Why C: Concentric perineurial layers with tight junctions surround each fascicle.

Why not: A) Loose tissue around individual axons. B) Outer dense sheath. D) Not a barrier layer.

Q10. The epineurium is:

A) Loose tissue around single axons
B) A myelin layer
C) A tight-junctional barrier
D) Dense irregular connective tissue binding fascicles, carrying the vasa nervorum

Answer: D · Difficulty: Medium · **LO 7.4**

Why D: It also contains adipose tissue.

Why not: A) Endoneurium. C) Perineurium. B) Schwann cell product.

Q11. Botulinum toxin causes flaccid paralysis by:

- A) Cleaving SNARE proteins, preventing ACh release at the NMJ
- B) Blocking acetylcholine receptors
- C) Inhibiting acetylcholinesterase
- D) Blocking glycine release

Answer: A · Difficulty: Medium · LO 7.5

Why A: No transmitter release means no muscle activation.

Why not: B) Curare-like agents. C) Would increase activity. D) Tetanus toxin.

Q12. Tetanus toxin causes spastic paralysis because it blocks release of:

- A) Acetylcholine at the NMJ B) Inhibitory glycine and GABA from spinal interneurons C) Noradrenaline D) Dopamine

Answer: B · Difficulty: Hard · LO 7.5

Why B: Loss of inhibition leaves motor neurons unopposed — the mirror image of botulism despite an identical molecular mechanism.

Why not: A) Botulinum. C, D) Not the target.

Q13. Myasthenia gravis results from autoantibodies against:

- A) Presynaptic Ca^{2+} channels B) MuSK only C) Postsynaptic nicotinic acetylcholine receptors D) Voltage-gated Na^+ channels

Answer: C · Difficulty: Medium · LO 7.5

Why C: Produces fatigable weakness that improves with anticholinesterases. (Anti-MuSK is a recognised minority subtype.)

Why not: A) Lambert-Eaton. B) A subset, not the classic answer. D) Not a target.

Q14. Lambert-Eaton syndrome differs clinically from myasthenia gravis because strength:

- A) Worsens with repeated use B) Is lost abruptly C) Is unaffected D) Improves with repeated use

Answer: D · Difficulty: Hard · LO 7.5

Why D: Presynaptic Ca^{2+} channel antibodies reduce release; repetitive activity accumulates Ca^{2+} and facilitates it.

Why not: A) Myasthenia. B, C) Incorrect.

Q15. Multiple sclerosis plaques are characterised by:

- A) Demyelination with relative axonal preservation, typically periventricular
- B) Axonal loss with myelin preservation
- C) Neuronal storage material
- D) Amyloid plaques

Answer: A · Difficulty: Medium · LO 7.6

Why A: The insulation, not the wire, is the primary target.

Why not: B) Reversed. C) Storage diseases. D) Alzheimer disease.

Q16. Guillain-Barré syndrome is frequently preceded by infection with:

A) Staphylococcus aureus B) Campylobacter jejuni C) Candida D) Influenza only

Answer: B · Difficulty: Medium · LO 7.6

Why B: Molecular mimicry between bacterial lipo-oligosaccharide and peripheral nerve gangliosides.

Why not: A, C) Not classically associated. D) Occasionally implicated but not the classic answer.

Q17. Ependymal cells:

A) Myelinate CNS axons B) Are phagocytic C) Line the ventricles and central canal, ciliated, aiding CSF circulation D) Form the blood-brain barrier

Answer: C · Difficulty: Easy · LO 7.2

Why C: Modified epithelium; choroid plexus epithelium produces CSF.

Why not: A) Oligodendrocytes. B) Microglia. D) Endothelial tight junctions with astrocytic induction.

Q18. A neuron with one axon and multiple dendrites is:

A) Unipolar B) Bipolar C) Pseudounipolar D) Multipolar

Answer: D · Difficulty: Easy · LO 7.1

Why D: The commonest type — motor neurons and most CNS neurons.

Why not: B) Retina, olfactory. C) Dorsal root ganglion sensory neurons.

Q19. Dorsal root ganglion neurons are:

A) Pseudounipolar, surrounded by satellite cells B) Multipolar C) Bipolar D) Anaxonic

Answer: A · Difficulty: Medium · LO 7.1

Why A: B single process bifurcates into peripheral and central branches; satellite cells encircle the somata.

Why not: B) Autonomic ganglia and CNS. C) Special sensory. D) Rare interneurons.

Q20. Wallerian degeneration refers to:

A) Degeneration of the proximal stump
B) Degeneration of the axon distal to injury, with myelin breakdown and macrophage clearance
C) Neuronal apoptosis
D) Demyelination without axonal injury

Answer: B · Difficulty: Medium · LO 7.6

Why B: The distal segment, separated from the perikaryon, cannot be maintained.

Why not: A) Limited retrograde change occurs but is not Wallerian. C, D) Different processes.

Q21. Chromatolysis in an injured neuron describes:

- A) Nuclear fragmentation
- B) Cytoplasmic calcification
- C) Dispersal of Nissl substance with peripheral nuclear displacement and cell swelling
- D) Myelin loss

Answer: C · Difficulty: Hard · LO 7.6

Why C: The reaction of the cell body to axonal injury, reflecting a shift to repair-oriented synthesis.

Why not: A) Karyorrhexis. B, D) Different phenomena.

Q22. The 20–30 nm gap at a chemical synapse is bridged by:

- A) Gap junction channels
- B) Tight junctions
- C) Direct cytoplasmic continuity
- D) Diffusing neurotransmitter

Answer: D · Difficulty: Easy · LO 7.5

Why D: Chemical transmission converts electrical to chemical signalling and back.

Why not: A, C) Electrical synapses. B) Not present.

Q23. On transverse section, a peripheral nerve is distinguished from tendon by:

- A) Fascicles of small round haloed profiles surrounded by a distinct perineurial ring
- B) Pink colour
- C) Flattened nuclei in rows
- D) Wavy parallel collagen

Answer: A · Difficulty: Medium · LO 7.10

Why A: The haloes are dissolved myelin around axons.

Why not: B) Both are pink. C, D) Both describe tendon.

Q24. Satellite cells in peripheral ganglia are functionally analogous to:

- A) Microglia
- B) Astrocytes/Schwann cells providing support and ionic regulation
- C) Ependyma
- D) Endothelium

Answer: B · Difficulty: Medium · LO 7.2

Why B: They encapsulate neuronal cell bodies and regulate their microenvironment.

Why not: A) Immune. C) Ventricular lining. D) Vascular.

Q25. Astrocyte end-feet contribute to the blood-brain barrier by:

- A) Forming the tight junctions themselves
- B) Phagocytosing pathogens
- C) Inducing and maintaining endothelial tight junctions
- D) Producing CSF

Answer: C · Difficulty: Hard · LO 7.2

Why C: The barrier resides in the endothelium, but astrocytes induce and sustain the phenotype — a frequently confused point.

Why not: A) Endothelial cells form them. B) Microglia. D) Choroid plexus.

Q26. The arrangement of grey and white matter is:

- A) Grey outside in both brain and spinal cord
- B) Grey inside in both
- C) Grey outside in the cerebrum and cerebellum, but inside in the spinal cord
- D) Randomly distributed

Answer: C · Difficulty: Easy · LO 7.6

Why C: Cortical grey lies superficially in brain; in the cord the grey forms a central butterfly surrounded by white funiculi. This inversion is the fastest orientation test on a CNS section.

Why not: A, B) Each is true of only one region. D) The arrangement is highly ordered.

Q27. A section shows a single row of very large flask-shaped neurons between an outer pale layer and an inner densely nucleated layer. This is:

- A) Cerebral cortex
- B) Dorsal root ganglion
- C) Spinal cord grey matter
- D) Cerebellar cortex, with Purkinje cells

Answer: D · Difficulty: Easy · LO 7.7

Why D: The single Purkinje cell row between molecular and granular layers is unique in the body.

Why not: A) Six indistinct layers with triangular pyramidal cells. C) Butterfly-shaped, no layering. B) Clustered round somata with satellite cells.

Q28. Purkinje cells are notable because they:

- A) Are the sole output of the cerebellar cortex and are inhibitory (GABAergic)
- B) Are the only excitatory output of the cerebellar cortex
- C) Receive no input
- D) Lie in the granular layer

Answer: A · Difficulty: Hard · LO 7.7

Why A: The entire cerebellar cortex computes onto one inhibitory output cell — a frequently examined point.

Why not: B) They are inhibitory, not excitatory. C) They receive parallel and climbing fibre input. D) They form their own layer.

Q29. An epidural haematoma is lens-shaped and arterial, whereas a subdural haematoma is crescent-shaped and venous, because:

- A) Blood clots differently in each space

- B) Dura is adherent to bone, so arterial blood must strip it away, while subdural venous blood spreads freely
- C) The arachnoid is impermeable
- D) CSF dilutes subdural blood

Answer: B · Difficulty: Hard · LO 7.8

Why B: The shape of each haemorrhage is a direct readout of meningeal anatomy and the vessel torn (middle meningeal artery versus bridging veins).

Why not: A) Clotting is the same. C, D) Neither determines the shape.

Q30. A ganglion shows clustered round neuronal somata, each fully encircled by satellite cells, with no synapses present. This is:

- A) An autonomic ganglion
- B) The choroid plexus
- C) A dorsal root ganglion
- D) A Meissner plexus

Answer: C · Difficulty: Medium · LO 7.9

Why C: Sensory information passes straight through a dorsal root ganglion, so no synapses occur there — the decisive difference from an autonomic ganglion.

Why not: A) Multipolar, scattered neurons with synapses and incomplete satellite rings. B) Cuboidal epithelium on vascular fronds. D) Enteric, within the gut wall.

Chapter 8 — Circulatory System

Q1. Elastic arteries store energy in systole and recoil in diastole because their media contains:

A) Mostly smooth muscle B) Cartilage C) Only collagen D) Concentric fenestrated elastic lamellae

Answer: D · Difficulty: Easy · LO 8.1

Why D: The Windkessel effect converts pulsatile ejection into more continuous flow.

Why not: A) Muscular arteries. B, C) Not media components of large arteries.

Q2. The chief site of peripheral vascular resistance is the:

A) Arteriole B) Muscular artery C) Aorta D) Vein

Answer: A · Difficulty: Easy · LO 8.1

Why A: Small radius and a 1–3 layer muscular media make arterioles the main regulators of blood pressure.

Why not: B, C) Conducting/distributing. D) Capacitance vessels.

Q3. On H&E, an artery is distinguished from its accompanying vein by:

A) Larger lumen B) A thick round muscular wall with a wavy internal elastic lamina C) Thicker adventitia D) Presence of valves

Answer: B · Difficulty: Easy · LO 8.5

Why B: The wavy IEL is the decisive feature; the vein collapses with a thin wall.

Why not: A, C, D are all venous characteristics.

Q4. Leukocyte diapedesis occurs predominantly at:

A) Arterioles B) Capillaries C) Postcapillary venules D) Large veins

Answer: C · Difficulty: Medium · LO 8.2

Why C: Their thin walls and adhesion-molecule expression make them the site of emigration and of histamine-induced leakage.

Why not: A, B, D are not the principal sites.

Q5. Fenestrated capillaries are characteristic of:

A) Brain B) Skeletal muscle C) Lung alveoli D) Endocrine glands and gut mucosa

Answer: D · Difficulty: Medium · LO 8.2

Why D: Pores permit rapid exchange of water and small solutes, including hormone entry into blood.

Why not: A, B, C) Continuous capillaries.

Q6. Discontinuous (sinusoidal) capillaries with gaps and absent basal lamina are found in:

A) Liver, spleen, bone marrow B) Muscle C) CNS D) Skin

Answer: A · Difficulty: Medium · LO 8.2

Why A: They permit passage of plasma proteins and whole cells.

Why not: B, C, D have continuous endothelium.

Q7. The glomerular capillary is fenestrated but unusual in that its fenestrations:

A) Have thick diaphragms B) Lack diaphragms C) Are absent D) Are intracellular

Answer: B · Difficulty: Hard · LO 8.2

Why B: Absent diaphragms maximise filtration; selectivity is provided by the GBM and podocyte slit diaphragms.

Why not: A) Most other fenestrated capillaries have them. C, D) Incorrect.

Q8. Weibel-Palade bodies store:

A) Histamine B) Renin C) von Willebrand factor and P-selectin D) Surfactant

Answer: C · Difficulty: Hard · LO 8.3

Why C: They allow endothelium to switch rapidly to a prothrombotic, pro-adhesive state.

Why not: A) Mast cells/basophils. B) Juxtaglomerular cells. D) Type II pneumocytes.

Q9. Endothelial nitric oxide is synthesised from:

A) Histidine B) Tyrosine C) Tryptophan D) L-arginine via eNOS

Answer: D · Difficulty: Medium · LO 8.3

Why D: NO causes vasodilation and inhibits platelet aggregation; its loss marks endothelial dysfunction.

Why not: A, B, C are precursors of other mediators.

Q10. The sequence of leukocyte emigration is:

A) Margination → rolling (selectins) → firm adhesion (integrins/ICAM) → transmigration

B) Adhesion → rolling → margination → transmigration

C) Transmigration → rolling → adhesion

D) Rolling → transmigration → adhesion

Answer: A · Difficulty: Medium · LO 8.3

Why A: Selectins mediate low-affinity rolling; integrins mediate firm arrest.

Why not: B, C, D misorder the cascade.

Q11. The earliest visible lesion of atherosclerosis is the:

A) Fibrous cap B) Fatty streak of foam cells C) Calcified plaque D) Aneurysm

Answer: B · Difficulty: Easy · LO 8.4

Why B: Macrophages take up oxidised LDL via scavenger receptors to form foam cells.

Why not: A) Later. C) Advanced. D) A complication.

Q12. Smooth muscle cells in an atheromatous plaque:

- A) Remain in the media
- B) Undergo apoptosis only
- C) Migrate into the intima, proliferate, and synthesise the collagen of the fibrous cap
- D) Become endothelium

Answer: C · Difficulty: Medium · LO 8.4

Why C: This phenotypic switch from contractile to synthetic builds the cap.

Why not: A, B, D are incorrect.

Q13. Acute myocardial infarction most often arises from a plaque that is:

- A) Severely stenotic and calcified
- B) Entirely endothelialised
- C) Small and fibrotic
- D) Lipid-rich with a thin fibrous cap, prone to rupture

Answer: D · Difficulty: Hard · LO 8.4

Why D: Composition, not degree of stenosis, determines the acute event — most infarcts follow rupture of non-critically stenotic plaques.

Why not: A) Causes stable angina. B, C) Low risk.

Q14. Plaques form preferentially at arterial bifurcations because:

- A) Disturbed, low laminar shear stress removes the atheroprotective endothelial phenotype
- B) Pressure is highest there
- C) There is no endothelium
- D) Cholesterol is denser there

Answer: A · Difficulty: Hard · LO 8.4

Why A: Laminar shear normally maintains protective gene expression.

Why not: B, C, D are incorrect.

Q15. Vasa vasorum are found in:

- A) Capillaries B) The adventitia and outer media of large vessels C) The intima only D) Arterioles

Answer: B · Difficulty: Medium · LO 8.1

Why B: The outer wall is too thick to be nourished by luminal diffusion.

Why not: A, D) Too small to need them. C) Nourished by the lumen.

Q16. Veins possess valves formed by folds of the:

- A) Media B) Adventitia C) Intima D) External elastic lamina

Answer: C · Difficulty: Medium · LO 8.1

Why C: Intimal folds prevent retrograde flow in low-pressure vessels.

Why not: A, B, D do not form valve cusps.

Q17. Approximately what proportion of the blood volume is held in the venous system?

A) 10% B) 30% C) 95% D) 70%

Answer: D · Difficulty: Medium · LO 8.1

Why D: Veins are the capacitance reservoir.

Why not: A, B, C are incorrect.

Q18. Pericytes are:

A) Contractile perivascular cells sharing the capillary basal lamina B) Endothelial cells C) Macrophages D) Fibroblasts of the adventitia

Answer: A · Difficulty: Medium · LO 8.2

Why A: They regulate capillary flow and contribute to repair and angiogenesis.

Why not: B, C, D are distinct cell types.

Q19. The cardiac skeleton electrically insulates atria from ventricles, which means:

A) Conduction is impossible

B) The AV node is the only normal conduction route, imposing the necessary delay

C) The atria contract after the ventricles

D) Purkinje fibres are unnecessary

Answer: B · Difficulty: Hard · LO 8.6

Why B: This anatomical fact underlies the physiology of AV delay.

Why not: A) Conduction proceeds via the AV node. C) Reversed. D) They are essential.

Q20. Purkinje fibres are:

A) Nerve fibres B) Collagen bundles C) Modified cardiac muscle cells, pale and glycogen-rich, in the subendocardium D) Smooth muscle

Answer: C · Difficulty: Medium · LO 8.6

Why C: "Fibre" here does not mean nerve — a classic trap.

Why not: A, B, D are incorrect.

Q21. The epicardium is equivalent to the:

A) Parietal pericardium B) Myocardium C) Endocardium D) Visceral pericardium — mesothelium over connective tissue

Answer: D · Difficulty: Easy · LO 8.6

Why D: It also carries the coronary vessels in adipose tissue.

Why not: A) The outer sac. B, C) Inner layers.

Q22. Lymphatic capillaries differ from blood capillaries by:

- A) Blind-ended origin, overlapping endothelium, anchoring filaments, and an incomplete basal lamina
- B) Having a complete basal lamina
- C) Containing erythrocytes
- D) Having a thick media

Answer: A · Difficulty: Medium · LO 8.7

Why A: This construction permits entry of interstitial fluid and large molecules.

Why not: B) Incomplete. C) Lymph is typically acellular apart from lymphocytes. D) None.

Q23. In the CNS, the blood-brain barrier depends principally on:

- A) Fenestrations B) Continuous endothelium with extensive tight junctions
- C) Sinusoids D) Pericyte absence

Answer: B · Difficulty: Easy · LO 8.2

Why B: Tight junctions restrict paracellular passage, maintaining a controlled neuronal environment.

Why not: A, C) Would defeat the barrier. D) Pericytes are present and contributory.

Q24. Endothelium maintains an antithrombotic surface through all EXCEPT:

- A) Thrombomodulin B) Prostacyclin C) Exposed subendothelial collagen D) Nitric oxide

Answer: C · Difficulty: Medium · LO 8.3

Why C: Exposure of collagen is precisely the prothrombotic trigger following injury.

Why not: A, B, D are all anticoagulant/antiplatelet mechanisms.

Q25. A vessel with a thick media of smooth muscle and a prominent, clearly visible internal elastic lamina, about 2 mm in diameter, is a:

- A) Elastic artery B) Vein C) Arteriole D) Muscular artery

Answer: D · Difficulty: Medium · LO 8.5

Why D: Muscular (distributing) arteries have the most conspicuous IEL.

Why not: A) Elastic lamellae throughout the media. C) Only 1-3 muscle layers. B) Thin wall, no prominent IEL.

Chapter 9 — Blood

Q1. Central pallor of the erythrocyte occupies normally about:

- A) One-third of the diameter B) One-tenth of the diameter C) Two-thirds D) None

Answer: A · Difficulty: Easy · LO 9.1

Why A: The biconcave shape makes the centre thinner, so less haemoglobin lies in the light path.

Why not: B, C, D indicate pathology or misreading.

Q2. Hypochromia with enlarged central pallor is typical of:

- A) Spherocytosis B) Iron deficiency and thalassaemia C) Megaloblastic anaemia D) Anaemia of chronic disease only

Answer: B · Difficulty: Easy · LO 9.2

Why B: Reduced haemoglobin content per cell increases the pale zone.

Why not: A) Loss of pallor. C) Macro-ovalocytes. D) Usually normochromic.

Q3. Spherocytes lack central pallor because:

- A) They contain excess haemoglobin
- B) They are immature
- C) Membrane loss has abolished the biconcave shape
- D) They are larger

Answer: C · Difficulty: Medium · LO 9.2

Why C: Loss of surface area relative to volume yields a sphere, which is uniformly thick.

Why not: A) Haemoglobin content is not increased overall. B) Not immaturity. D) They are smaller in diameter.

Q4. Target cells arise when:

- A) Haemoglobin content exceeds membrane
- B) Iron is excessive
- C) The cell is dehydrated
- D) The ratio of membrane surface area to haemoglobin content is increased

Answer: D · Difficulty: Hard · LO 9.2

Why D: Excess membrane relative to contents allows a bell shape that appears as a central dot in a ring of pallor — hence their presence in thalassaemia, iron deficiency, and liver disease.

Why not: A) Reversed. B, C) Not the mechanism.

Q5. The most numerous leukocyte in normal adult blood is the:

- A) Neutrophil B) Lymphocyte C) Monocyte D) Eosinophil

Answer: A · Difficulty: Easy · LO 9.3

Why A: 50–70%, ahead of lymphocytes (20–40%).

Why not: B) Second. C, D) Minor populations.

Q6. The largest circulating leukocyte, with an indented nucleus and grey-blue cytoplasm, is the:

A) Neutrophil B) Monocyte C) Large lymphocyte D) Basophil

Answer: B · Difficulty: Easy · LO 9.3

Why B: Kidney/horseshoe nucleus with lacy chromatin and often vacuoles.

Why not: A) Multilobed. C) Round dense nucleus, deeply basophilic cytoplasm. D) Obscured by dark granules.

Q7. Hypersegmented neutrophils (>5 lobes) suggest:

A) Bacterial infection B) Allergy C) Megaloblastic anaemia D) Chronic myeloid leukaemia

Answer: C · Difficulty: Medium · LO 9.4

Why C: Impaired DNA synthesis from B12/folate deficiency.

Why not: A) Left shift with band forms. B) Eosinophilia. D) Left shift with basophilia.

Q8. Eosinophilia is classically associated with:

A) Pyogenic bacterial infection
B) Aplastic anaemia
C) Viral hepatitis
D) Helminth (parasitic worm) infection, allergy, and drug reactions

Answer: D · Difficulty: Easy · LO 9.4

Why D: Major basic protein kills helminths extracellularly, as they are too large to phagocytose.

Why not: A) Neutrophilia. C) Lymphocytosis. B) Pancytopenia.

Q9. Chronic granulomatous disease results from a defect in:

A) NADPH oxidase B) Myeloperoxidase only C) Lysozyme D) Defensins

Answer: A · Difficulty: Hard · LO 9.4

Why A: No respiratory burst, so catalase-positive organisms survive; diagnosed by dihydrorhodamine flow cytometry (historically NBT).

Why not: B) B separate, milder deficiency. C, D) Not the primary defect.

Q10. Basophils and mast cells are best distinguished by the fact that:

A) They are the same cell
B) Basophils circulate and are terminally differentiated; mast cells mature in tissue and are long-lived
C) Only mast cells bind IgE
D) Only basophils contain histamine

Answer: B · Difficulty: Hard · LO 9.3

Why B: Separate lineage trajectories despite similar granule contents.

Why not: A) Distinct cells. C, D) Both bind IgE and contain histamine.

Q11. B, T, and NK lymphocytes are distinguished by:

A) Size on a smear B) Nuclear shape C) Immunophenotyping (CD19/20, CD3, CD56) D) Granule colour

Answer: C · Difficulty: Medium · LO 9.3

Why C: They are morphologically indistinguishable — an important limit of morphology.

Why not: A, B, D cannot reliably separate them.

Q12. Atypical lymphocytes of infectious mononucleosis are characteristically:

A) Small with scant cytoplasm

B) Granular and red

C) Multilobed

D) Large with abundant basophilic cytoplasm indented by adjacent red cells

Answer: D · Difficulty: Medium · LO 9.4

Why D: These are reactive CD8⁺ T cells responding to EBV-infected D cells.

Why not: A) Normal small lymphocyte. C) Neutrophil. B) Eosinophil.

Q13. Platelets are:

A) Anucleate cytoplasmic fragments of megakaryocytes

B) Small nucleated cells

C) Erythrocyte fragments

D) Lymphocyte derivatives

Answer: A · Difficulty: Easy · LO 9.5

Why A: 2–4 μm , with a central granulomere and peripheral hyalomere.

Why not: B, C, D are incorrect.

Q14. Platelet α -granules contain:

A) ADP and serotonin B) Fibrinogen, von Willebrand factor and PDGF C) Histamine D) Myeloperoxidase

Answer: B · Difficulty: Hard · LO 9.5

Why B: Dense granules hold ADP, ATP, calcium, and serotonin.

Why not: A) Dense granules. C) Mast cells. D) Neutrophils.

Q15. Heinz bodies and bite cells indicate:

A) Iron deficiency B) Sickle cell disease C) G6PD deficiency with oxidative haemolysis D) Thalassaemia

Answer: C · Difficulty: Hard · LO 9.6

Why C: Denatured haemoglobin precipitates and is removed by splenic macrophages, leaving bite-shaped defects.

Why not: A) Pencil cells. B) Sickle forms. D) Target cells.

Q16. Hereditary spherocytosis most often involves defects of:

A) Haemoglobin B) Pyruvate kinase C) G6PD D) Spectrin, ankyrin or band 3 of the membrane skeleton

Answer: D · Difficulty: Medium · LO 9.6

Why D: Vertical linkage failure causes membrane loss and sphering.

Why not: A) Haemoglobinopathies. B, C) Enzymopathies.

Q17. Splenectomy benefits hereditary spherocytosis because:

A) It removes the site where rigid spherocytes are trapped and destroyed
B) It increases red cell production
C) It corrects the membrane defect
D) It raises erythropoietin

Answer: A · Difficulty: Medium · LO 9.6

Why A: The cells remain spherocytic but survive longer.

Why not: B, C, D are incorrect.

Q18. Howell-Jolly bodies in the peripheral blood indicate:

A) Iron overload B) Absent or non-functioning spleen C) Lead poisoning D) Active infection

Answer: B · Difficulty: Medium · LO 9.6

Why B: Nuclear remnants normally "pitted" out by splenic macrophages persist.

Why not: A) Siderocytes. C) Basophilic stippling. D) Toxic granulation.

Q19. The erythrocyte depends for ATP on:

A) Oxidative phosphorylation B) β -oxidation C) Anaerobic glycolysis D) Ketone bodies

Answer: C · Difficulty: Medium · LO 9.1

Why C: It has no mitochondria; the hexose monophosphate shunt supplies NADPH for antioxidant defence.

Why not: A, B, D all require mitochondria.

Q20. The Mentzer index ($MCV \div RBC \text{ count}$) below 13 favours:

A) Iron deficiency B) Aplastic anaemia C) Megaloblastic anaemia D) Thalassaemia trait

Answer: D · Difficulty: Hard · LO 9.7

Why D: Thalassaemia trait shows a preserved/high red cell count with disproportionately low MCV.

Why not: A) Index above 13. B, C) Not microcytic.

Q21. A "left shift" refers to:

A) Increased band forms and immature granulocytes B) Hypersegmentation C) Lymphocytosis D) Reticulocytosis

Answer: A · Difficulty: Easy · LO 9.4

Why A: Reflects accelerated marrow release, typically in bacterial infection.

Why not: B) Megaloblastic. C) Viral. D) Erythroid response.

Q22. Polychromasia on a Romanowsky-stained film corresponds to:

A) Spherocytes B) Reticulocytes C) Target cells D) Schistocytes

Answer: B · Difficulty: Medium · LO 9.7

Why B: Residual ribosomal RNA imparts a blue-grey tinge to these slightly large cells.

Why not: A, C, D have different mechanisms and appearances.

Q23. Schistocytes (fragmented red cells) suggest:

A) Nutritional deficiency B) Thalassaemia C) Microangiopathic haemolysis (e.g. DIC, TTP, mechanical valve) D) Spherocytosis

Answer: C · Difficulty: Medium · LO 9.6

Why C: Mechanical shearing across fibrin strands or prosthetic surfaces.

Why not: A, B, D produce different morphologies.

Q24. In sickle cell disease, the primary molecular event is:

- A) Reduced globin chain synthesis
- B) Membrane skeleton defect
- C) Oxidation of haemoglobin
- D) Polymerisation of deoxygenated HbS into rigid fibres

Answer: D · Difficulty: Medium · LO 9.6

Why D: Polymerisation deforms the cell, producing haemolysis and vaso-occlusion.

Why not: A) Thalassaemia. C) G6PD. B) Spherocytosis.

Q25. A small lymphocyte's nucleus is approximately the size of:

A) One erythrocyte B) A platelet C) Three erythrocytes D) A megakaryocyte

Answer: A · Difficulty: Easy · LO 9.3

Why A: This equivalence is the practical scale reference on any film.

Why not: B) Much smaller. C) Would indicate a large cell. D) Vastly larger.

Chapter 10 — Hemopoiesis

Q1. The last mitotically capable stage of erythroid maturation is the:

A) Proerythroblast B) Polychromatophilic erythroblast C) Basophilic erythroblast D) Orthochromatophilic erythroblast

Answer: B · Difficulty: Medium · LO 10.2

Why B: Beyond this stage the nucleus is too condensed to divide.

Why not: A, C) Earlier dividing stages. D) Post-mitotic, about to extrude its nucleus.

Q2. As a haemopoietic cell matures, the general rule is that:

A) Cell size increases and nucleoli appear
B) Cytoplasm becomes more basophilic
C) Cell size decreases, chromatin condenses, nucleoli disappear, N:C ratio falls
D) The nucleus enlarges

Answer: C · Difficulty: Easy · LO 10.2

Why C: This single rule allows reconstruction of every lineage.

Why not: A, B, D describe the reverse direction.

Q3. The reticulocyte is identified by supravital staining with new methylene blue because it contains:

A) Residual DNA B) Denatured haemoglobin C) Iron granules D) Residual ribosomal RNA

Answer: D · Difficulty: Medium · LO 10.2

Why D: The precipitated RNA forms the reticulum that gives the cell its name.

Why not: A) The nucleus is already extruded. C) Siderocytes. B) Heinz bodies.

Q4. A raised reticulocyte count in an anaemic patient indicates:

A) An appropriate marrow response, as in haemolysis or blood loss B) Marrow failure C) Iron deficiency D) Aplastic anaemia

Answer: A · Difficulty: Medium · LO 10.2

Why A: It shows the marrow can respond when substrate and stimulus are present.

Why not: B, C, D would show an inappropriately low count.

Q5. In granulopoiesis, the last stage capable of mitosis is the:

A) Myeloblast B) Myelocyte C) Promyelocyte D) Metamyelocyte

Answer: B · Difficulty: Medium · LO 10.3

Why B: Specific granules appear at this stage, which is also the final dividing one.

Why not: A, C) Earlier dividing stages. D) Post-mitotic with an indented nucleus.

Q6. Primary (azurophilic) granules first appear at the stage of the:

A) Myeloblast B) Myelocyte C) Promyelocyte D) Band cell

Answer: C · Difficulty: Medium · LO 10.3

Why C: They are lysosomes containing myeloperoxidase, elastase, and defensins.

Why not: A) No granules yet. B) Specific granules. D) Too late.

Q7. Lineage-specific (neutrophil/eosinophil/basophil) identity becomes morphologically apparent with the appearance of:

A) Primary granules B) The nuclear indentation C) Nucleoli D) Secondary (specific) granules

Answer: D · Difficulty: Medium · LO 10.3

Why D: Specific granules define the lineage.

Why not: A) Common to all. B, C) Maturation markers, not lineage markers.

Q8. Erythropoietin is produced by the:

A) Kidney peritubular interstitial cells in response to hypoxia B) Liver C) Bone marrow stroma D) Spleen

Answer: A · Difficulty: Easy · LO 10.4

Why A: This explains the anaemia of chronic kidney disease and the use of recombinant EPO.

Why not: B) Produces thrombopoietin (and fetal EPO). C, D) Not the main source.

Q9. Thrombopoietin is produced mainly by the:

A) Kidney B) Liver C) Spleen D) Thymus

Answer: B · Difficulty: Medium · LO 10.4

Why B: Constitutive hepatic production regulates megakaryocyte maturation.

Why not: A) EPO. C, D) Not significant sources.

Q10. Megakaryocytes release platelets by:

A) Complete cell fragmentation in the sinusoidal lumen

B) Apoptosis in the cords

C) Extending proplatelet processes through sinusoidal endothelium and shedding platelets into the blood

D) Budding into lymphatics

Answer: C · Difficulty: Hard · LO 10.3

Why C: Their position against the sinusoid wall is functionally essential.

Why not: A, B, D are incorrect mechanisms.

Q11. Erythroblastic islands are organised around a central:

A) Megakaryocyte B) Endothelial cell C) Fibroblast D) Macrophage supplying iron and phagocytosing extruded nuclei

Answer: D · Difficulty: Hard · LO 10.3

Why D: The "nurse cell" arrangement couples iron handling to erythropoiesis.

Why not: A, B, C do not serve this role.

Q12. Only mature cells normally enter the blood from marrow because:

- A) Sinusoidal endothelium selects by deformability and surface receptors
- B) Immature cells are destroyed
- C) There is a basement membrane barrier
- D) Immature cells lack haemoglobin

Answer: A · Difficulty: Hard · LO 10.5

Why A: The barrier is selective by maturity, which is why circulating blasts are pathological.

Why not: B) Not destroyed. C) The basal lamina is deficient. D) Irrelevant to transit.

Q13. In the adult, active red marrow is found principally in the:

- A) Long bones of the limbs
- B) Vertebrae, sternum, ribs, pelvis, skull, proximal femur and humerus
- C) Phalanges
- D) Skull only

Answer: B · Difficulty: Easy · LO 10.5

Why B: Hence the posterior iliac crest as the standard biopsy site.

Why not: A, C) Converted to yellow marrow. D) Incomplete.

Q14. Extramedullary haemopoiesis in chronic severe anaemia occurs in the:

- A) Kidney and lung B) Brain C) Liver and spleen D) Skin

Answer: C · Difficulty: Medium · LO 10.5

Why C: The fetal sites resume activity, producing hepatosplenomegaly.

Why not: A, B, D are not haemopoietic sites.

Q15. The "hair-on-end" skull appearance in severe thalassaemia results from:

- A) Infection B) Calcification C) Tumour D) Marrow cavity expansion thinning the cortical bone

Answer: D · Difficulty: Hard · LO 10.5

Why D: Massive erythroid hyperplasia expands the diploë — a histological cause of a radiological sign.

Why not: A, B, C are unrelated.

Q16. The normal adult myeloid-to-erythroid ratio is approximately:

- A) 2:1 to 4:1 B) 1:1 C) 1:3 D) 10:1

Answer: A · Difficulty: Medium · LO 10.5

Why A: Granulocytes live hours to days; erythrocytes last 120 days, so more granulocytic precursors are required.

Why not: B, C, D are incorrect.

Q17. A useful estimate of normal adult marrow cellularity is:

A) Always 100% B) Approximately 100 minus the patient's age C) Always 50% D) Age plus 20

Answer: B · Difficulty: Medium · LO 10.6

Why B: Marrow becomes progressively fattier with age.

Why not: A, C, D do not account for age.

Q18. The haemopoietic stem cell is characterised by:

A) Lineage commitment B) Inability to divide C) Self-renewal plus multilineage differentiation capacity, CD34+ D) Restriction to the myeloid series

Answer: C · Difficulty: Medium · LO 10.1

Why C: Self-renewal is the defining property.

Why not: A, D) Progenitors. B) False.

Q19. Common lymphoid progenitors give rise to:

A) Erythrocytes and platelets B) Monocytes C) Neutrophils D) B, T and NK cells

Answer: D · Difficulty: Easy · LO 10.1

Why D: CMP yields the myeloid lineages; CLP the lymphoid.

Why not: A, B, C derive from the common myeloid progenitor.

Q20. G-CSF is used clinically to:

A) Accelerate neutrophil recovery after chemotherapy B) Raise platelet count C) Treat anaemia D) Suppress immunity

Answer: A · Difficulty: Easy · LO 10.4

Why A: Filgrastim shortens the period of neutropenia.

Why not: B) TPO agonists. C) EPO. D) Not immunosuppressive.

Q21. Fetal haemopoiesis occurs in the following sequence:

A) Bone marrow → liver → yolk sac
B) Yolk sac → liver (and spleen) → bone marrow
C) Liver → yolk sac → spleen
D) Spleen → yolk sac → marrow

Answer: B · Difficulty: Medium · LO 10.5

Why B: Marrow becomes dominant from about 20 weeks onward.

Why not: A, C, D misorder the sequence.

Q22. The megakaryocyte is recognised in marrow by:

A) A small round nucleus

- B) Multiple separate nuclei
- C) Very large size with a single enormous multilobed convoluted nucleus
- D) Bilobed nucleus with red granules

Answer: C · Difficulty: Easy · LO 10.6

Why C: Endomitosis produces polyploidy (up to 64N) within one nucleus.

Why not: A) Too small. B) It is one polyploid nucleus, not many. D) Eosinophil.

Q23. Absence of megakaryocytes together with loss of all lineages in a markedly hypocellular marrow suggests:

- A) Immune thrombocytopenia
- B) Iron deficiency
- C) Myelodysplasia
- D) Aplastic anaemia

Answer: D · Difficulty: Medium · LO 10.6

Why D: Trilineage failure with fatty replacement.

Why not: A) Megakaryocytes are increased. C) Dysplastic but usually cellular. B) Erythroid changes only.

Q24. In immune thrombocytopenia the marrow typically shows:

- A) Normal or increased megakaryocytes
- B) Absent megakaryocytes
- C) Fibrosis
- D) Blasts

Answer: A · Difficulty: Hard · LO 10.6

Why A: The defect is peripheral destruction, so the marrow compensates.

Why not: B) Aplasia. C) Myelofibrosis. D) Leukaemia.

Q25. Reconversion of yellow marrow to red marrow occurs in:

- A) Normal ageing
- B) Chronic severe anaemia or increased haemopoietic demand
- C) Obesity
- D) Dehydration

Answer: B · Difficulty: Medium · LO 10.5

Why B: The process is reversible under sustained stimulus.

Why not: A) Drives the opposite change. C, D) Unrelated.

Chapter 11 — Immune System and Lymphoid Organs

Q1. The decisive histological feature of the thymus is the:

A) Germinal centre B) Subcapsular sinus C) Hassall's corpuscle D) Central arteriole

Answer: C · Difficulty: Easy · LO 11.3

Why C: Whorled keratinised medullary epithelial cells found in no other organ.

Why not: A) Secondary lymphoid organs. B) Lymph node. D) Spleen.

Q2. The thymus lacks afferent lymphatics because it:

A) Is too small
B) Is not a lymphoid organ
C) Has no blood supply
D) Must exclude foreign antigen while educating T cells against self

Answer: D · Difficulty: Hard · LO 11.3

Why D: Architecture follows the antigen's route; here antigen must be kept out.

Why not: A, B, C are incorrect.

Q3. Positive selection of thymocytes occurs in the ____ and selects for ____:

A) Cortex; ability to recognise self-MHC B) Medulla; low self-affinity C) Cortex; strong self-peptide binding D) Medulla; MHC restriction

Answer: A · Difficulty: Medium · LO 11.3

Why A: Cells that cannot recognise self-MHC die by neglect; this establishes MHC restriction.

Why not: B, D) Negative selection occurs in the medulla. C) That triggers deletion.

Q4. AIRE enables medullary thymic epithelial cells to:

A) Produce antibodies
B) Express tissue-specific self-antigens for negative selection
C) Phagocytose thymocytes
D) Secrete thymosin

Answer: B · Difficulty: Hard · LO 11.3

Why B: Its mutation causes APECED/APS-1, proving negative selection prevents autoimmunity.

Why not: A, C, D are other functions of cells.

Q5. Germinal centres are found in:

A) Primary lymphoid organs B) The thymic cortex normally C) Secondary lymphoid organs following antigen exposure D) Bone marrow

Answer: C · Difficulty: Medium · LO 11.2

Why C: They are sites of somatic hypermutation, affinity maturation and class switching.

Why not: A, B, D) Their presence in the thymus is abnormal (e.g. myasthenia gravis).

Q6. High endothelial venules are located in the lymph node:

A) Subcapsular sinus B) Medullary sinus C) Germinal centre D) Paracortex

Answer: D · Difficulty: Medium · LO 11.4

Why D: Their plump cuboidal endothelium allows circulating lymphocytes to enter the node from blood.

Why not: A, B, C do not contain HEVs.

Q7. The paracortex of a lymph node is populated chiefly by:

A) T cells B) B cells C) Plasma cells D) Macrophages only

Answer: A · Difficulty: Easy · LO 11.4

Why A: Follicles (cortex) are B-cell zones; medullary cords contain plasma cells.

Why not: B) Cortex. C) Medulla. D) Sinuses.

Q8. Marked paracortical expansion in a reactive lymph node suggests:

A) Bacterial infection with a humoral response
B) A viral infection or drug reaction (T-cell response)
C) Metastatic carcinoma
D) Amyloidosis

Answer: B · Difficulty: Hard · LO 11.4

Why B: Reading the pattern of reaction reveals the type of immune response.

Why not: A) Produces follicular hyperplasia. C) Subcapsular sinus deposits. D) Congo red positive deposits.

Q9. Florid follicular hyperplasia indicates a predominantly:

A) T-cell response B) Granulomatous response C) B-cell/humoral response
D) Neoplastic process

Answer: C · Difficulty: Medium · LO 11.4

Why C: Enlarged germinal centres reflect B-cell activation.

Why not: A) Paracortical. B) Granulomas. D) Not necessarily.

Q10. The spleen's white pulp is recognised by the presence of:

A) A subcapsular sinus B) Villi C) Hassall's corpuscles D) An eccentric central arteriole within the lymphoid nodule

Answer: D · Difficulty: Medium · LO 11.5

Why D: A lymph node follicle never contains an arteriole.

Why not: A) Lymph node. C) Thymus. B) Intestine.

Q11. The periarteriolar lymphoid sheath (PALS) contains mainly:

A) T cells B) B cells C) Plasma cells D) Erythrocytes

Answer: A · Difficulty: Medium · LO 11.5

Why A: B-cell follicles lie eccentrically alongside the PALS.

Why not: B) Follicles. C) Cords. D) Red pulp.

Q12. The open circulation of the spleen is functionally essential because it:

A) Speeds blood flow

B) Forces red cells to squeeze between stave cells, filtering out rigid or abnormal cells

C) Prevents clotting

D) Oxygenates the blood

Answer: B · Difficulty: Hard · LO 11.5

Why B: This mechanical test underlies splenomegaly in haemolytic disease and the benefit of splenectomy.

Why not: A, C, D are incorrect.

Q13. Post-splenectomy patients are at risk of overwhelming infection with:

A) Viruses only

B) Fungi only

C) Encapsulated bacteria — pneumococcus, Hib, meningococcus

D) Helminths

Answer: C · Difficulty: Medium · LO 11.5

Why C: The spleen is the principal site of antibody responses to polysaccharide capsular antigens.

Why not: A, B, D are not the characteristic risk.

Q14. The cords of Billroth are located in the:

A) White pulp B) Capsule C) Marginal zone D) Red pulp

Answer: D · Difficulty: Medium · LO 11.5

Why D: A reticular meshwork with macrophages between sinusoids.

Why not: A, C) Lymphoid compartments. B) Fibrous covering.

Q15. M cells of Peyer's patches function to:

A) Transcytose luminal antigen to underlying immune cells

B) Secrete IgA

C) Absorb nutrients

D) Produce mucus

Answer: A · Difficulty: Medium · LO 11.6

Why A: Their basolateral pocket holds lymphocytes and dendritic cells.

Why not: B) Plasma cells. C) Enterocytes. D) Goblet cells.

Q16. Salmonella and Shigella exploit M cells because these cells:

A) Lack a membrane B) Provide a transcytotic portal across the epithelium

C) Secrete nutrients D) Are immunodeficient

Answer: B · Difficulty: Hard · LO 11.6

Why B: Any gateway can be subverted by pathogens.

Why not: A, C, D are incorrect.

Q17. Secretory IgA resists luminal proteolysis because of its:

A) Dimeric structure alone B) J chain only C) Retained secretory component D) Glycosylation of the Fab region

Answer: C · Difficulty: Hard · LO 11.6

Why C: The cleaved remnant of the polymeric Ig receptor protects the molecule.

Why not: A, B) Necessary for transport but not for protection. D) Not the mechanism.

Q18. Tingible-body macrophages produce the "starry sky" appearance in germinal centres because they:

A) Are pigmented B) Store iron C) Are infected D) Contain apoptotic debris of B cells failing selection

Answer: D · Difficulty: Medium · LO 11.4

Why D: They appear as pale cells against a dark background of proliferating lymphocytes.

Why not: A, B, C are incorrect.

Q19. The mantle zone of a secondary follicle consists of:

A) Compressed naive B cells displaced by the germinal centre B) Proliferating blasts C) T cells D) Macrophages

Answer: A · Difficulty: Medium · LO 11.4

Why A: It is the remnant of the primary follicle.

Why not: B) Germinal centre. C) Paracortex. D) Sinuses.

Q20. In HIV infection, the lymph node characteristically shows:

A) Stable architecture
B) Progressive paracortical CD4+ depletion, with follicles first florid then involuting
C) Granulomas only
D) Fibrosis from the outset

Answer: B · Difficulty: Hard · LO 11.4

Why B: The node's histology tracks the course of the disease.

Why not: A, C, D do not describe the typical evolution.

Q21. Lymph flows through a node in the order:

A) Efferent → medullary → subcapsular → afferent
B) Afferent → medulla → cortex → efferent
C) Afferent → subcapsular sinus → cortical/trabecular sinuses → medullary sinuses → efferent
D) Hilum → capsule → cortex

Answer: C · Difficulty: Medium · LO 11.4

Why C: Multiple afferents enter the convex surface; a single efferent leaves at the hilum.

Why not: A, B, D misorder the flow.

Q22. Primary lymphoid organs are:

A) Lymph node and spleen B) Spleen and thymus C) Tonsil and appendix D) Bone marrow and thymus

Answer: D · Difficulty: Easy · LO 11.2

Why D: Sites of lymphocyte generation and education without foreign antigen.

Why not: A, C) Secondary. B) Mixed.

Q23. Thymic involution after puberty is characterised by:

A) Replacement of parenchyma by adipose tissue, with Hassall's corpuscles persisting

B) Complete disappearance

C) Fibrosis with calcification

D) Conversion to lymph node

Answer: A · Difficulty: Medium · LO 11.3

Why A: This is how an adult thymus remains identifiable.

Why not: B, C, D are incorrect.

Q24. The palatine tonsil is distinguished from a lymph node by:

A) Having a complete capsule and sinuses

B) Being covered by stratified squamous epithelium with deep branching crypts and lacking afferent lymphatics

C) Containing Hassall's corpuscles

D) Having a central arteriole

Answer: B · Difficulty: Medium · LO 11.6

Why B: MALT is unencapsulated and directly beneath an epithelium.

Why not: A) Lymph node. C) Thymus. D) Spleen.

Q25. A section shows lymphoid follicles beneath an intestinal epithelium containing villi, with no capsule. This is:

A) Lymph node B) Spleen C) Peyer's patch D) Thymus

Answer: C · Difficulty: Medium · LO 11.6

Why C: Unencapsulated MALT in the ileal mucosa/submucosa with M cells in the overlying epithelium.

Why not: A) Encapsulated with sinuses. B) Central arterioles and red pulp. D) Hassall's corpuscles.

Q26. Which correctly pairs a lymphoid organ with its class?

A) Thymus — secondary

B) Lymph node — primary

C) Spleen — primary

D) Bone marrow and thymus — primary; lymph node, spleen and MALT — secondary

Answer: D · Difficulty: Easy · LO 11.1

Why D: Primary organs generate and educate lymphocytes without foreign antigen; secondary organs are where mature lymphocytes meet antigen, and only they contain germinal centres.

Why not: A, B, C all misassign an organ where antigen-driven activation occurs to the primary class.

Chapter 12 — Digestive System

Q1. Submucosal glands are found in only two regions of the digestive tract:

A) Stomach and colon B) Ileum and appendix C) Jejunum and ileum D) Oesophagus and duodenum

Answer: D · Difficulty: Medium · LO 12.1

Why D: Oesophageal mucous glands and duodenal Brunner glands — one of the most reliable identification clues.

Why not: A, B, C have no submucosal glands.

Q2. The myenteric (Auerbach) plexus lies:

A) Between the inner circular and outer longitudinal muscle layers B) In the lamina propria C) In the submucosa D) In the serosa

Answer: A · Difficulty: Easy · LO 12.2

Why A: Its position next to the muscle it governs reflects its role in motility.

Why not: B, C) Submucosal plexus territory. D) Not a plexus site.

Q3. The submucosal (Meissner) plexus principally controls:

A) Motility B) Secretion and local blood flow C) Swallowing D) Vomiting

Answer: B · Difficulty: Medium · LO 12.2

Why B: Position predicts function — it lies adjacent to the mucosa.

Why not: A) Myenteric plexus. C, D) Central reflexes.

Q4. In Hirschsprung disease, the diseased segment is:

A) The dilated proximal segment B) The entire colon always C) The narrow distal aganglionic segment D) The ileum

Answer: C · Difficulty: Hard · LO 12.2

Why C: The aganglionic segment remains tonically contracted; proximal dilatation is secondary — a classic trap.

Why not: A) Secondary change. B, D) Usually rectosigmoid.

Q5. The definitive diagnosis of Hirschsprung disease rests on:

A) Barium enema alone

B) Manometry only

C) Serum markers

D) Rectal suction biopsy showing absent ganglion cells with hypertrophied nerve trunks

Answer: D · Difficulty: Medium · LO 12.2

Why D: Histology is definitive.

Why not: A, B, C are supportive at best.

Q6. Parietal cells are recognised by:

- A) Large round/pyramidal shape with intensely eosinophilic cytoplasm and a central round nucleus
- B) Basophilic basal cytoplasm with apical granules
- C) Pale foamy cytoplasm
- D) Flattened basal nucleus

Answer: A · Difficulty: Easy · LO 12.3

Why A: Their abundant mitochondria provide ATP for proton pumping.

Why not: B) Chief cells. C, D) Mucous cells.

Q7. Parietal cells secrete:

- A) Pepsinogen and lipase B) HCl and intrinsic factor C) Gastrin D) Mucus and bicarbonate

Answer: B · Difficulty: Easy · LO 12.3

Why B: One cell type, two products — and two clinical syndromes if lost.

Why not: A) Chief cells. C) G cells. D) Surface mucous cells.

Q8. Autoimmune destruction of parietal cells causes:

- A) Iron deficiency only
- B) Coeliac disease
- C) Achlorhydria plus B12 deficiency (pernicious anaemia)
- D) Peptic ulcer

Answer: C · Difficulty: Medium · LO 12.3

Why C: Loss of intrinsic factor impairs ileal B12 absorption, causing megaloblastic anaemia and subacute combined degeneration.

Why not: A) A secondary consequence at most. B, D) Different mechanisms.

Q9. The three converging stimuli for gastric acid secretion are:

- A) Secretin, CCK, VIP B) Insulin, glucagon, GIP C) Somatostatin, gastrin, motilin D) Gastrin, histamine, acetylcholine

Answer: D · Difficulty: Medium · LO 12.3

Why D: This is why PPIs, acting on the final common effector, outperform H₂ blockers.

Why not: A, B, C are not the acid-secretory triad.

Q10. Tubulovesicles in the resting parietal cell contain:

- A) H⁺/K⁺-ATPase awaiting insertion into the canalicular membrane B) Pepsinogen C) Mucin D) Intrinsic factor only

Answer: A · Difficulty: Hard · LO 12.3

Why A: On stimulation they fuse with the canaliculus, expanding surface area and inserting pumps.

Why not: B) Chief cells. C) Mucous cells. D) Secreted separately.

Q11. Brunner glands are located in the:

A) Lamina propria of the jejunum B) Submucosa of the duodenum C) Muscularis of the stomach D) Mucosa of the ileum

Answer: B · Difficulty: Easy · LO 12.4

Why B: They secrete alkaline mucus neutralising acidic chyme — position, function and identification coincide.

Why not: A, C, D are incorrect.

Q12. Peyer's patches are most prominent in the:

A) Duodenum B) Jejunum C) Ileum D) Colon

Answer: C · Difficulty: Easy · LO 12.4

Why C: Lymphoid aggregates spanning mucosa into submucosa are the decisive ileal feature.

Why not: A, B) Absent or sparse. D) Different lymphoid pattern.

Q13. The decisive difference between colon and small intestine on section is:

A) Goblet cell number B) Epithelial type C) Presence of a muscularis mucosae D) Absence of villi in the colon

Answer: D · Difficulty: Easy · LO 12.5

Why D: Colonic mucosa presents a flat surface pierced by straight crypts.

Why not: A) Suggestive but not decisive. C) Both have one. B) Both are simple columnar.

Q14. Villi are distinguished from obliquely cut crypts by the fact that villi:

A) Project above the surface and contain a lamina propria core with a central lacteal

B) Contain goblet cells

C) Are lined by squamous cells

D) Lie below the muscularis mucosae

Answer: A · Difficulty: Medium · LO 12.5

Why A: Ask which side of the muscularis mucosae the structure lies on.

Why not: B) Both do. C) Neither. D) Neither.

Q15. Taeniae coli represent:

A) Thickened circular muscle B) Three bands of the outer longitudinal muscle layer C) Mucosal folds D) Serosal tags

Answer: B · Difficulty: Medium · LO 12.5

Why B: Their tone produces haustra.

Why not: A) Circular layer is complete. C) Plicae. D) Appendices epiploicae.

Q16. The oesophagus in the thorax is covered externally by:

A) Serosa B) Mesothelium C) Adventitia D) Peritoneum

Answer: C · Difficulty: Medium · LO 12.3

Why C: Being fixed and largely retroperitoneal/mediastinal, it lacks a serosa — a frequently tested point.

Why not: A, B, D imply a free peritoneal surface.

Q17. Barrett oesophagus is diagnosed by the presence above the gastro-oesophageal junction of:

- A) Squamous epithelium B) Brunner glands C) Parietal cells D) Intestinal-type columnar epithelium with goblet cells

Answer: D · Difficulty: Medium · LO 12.6

Why D: Goblet cells are the decisive finding of intestinal metaplasia; it carries adenocarcinoma risk.

Why not: A) Normal. B, C) Not diagnostic of Barrett.

Q18. Helicobacter pylori survives gastric acid principally by:

- A) Producing urease to generate ammonia B) Forming spores C) Invading parietal cells D) Coating itself in mucin

Answer: A · Difficulty: Medium · LO 12.6

Why A: Ammonia neutralises its immediate microenvironment within the mucus layer.

Why not: B, C) Not its biology. D) It lives in mucus but that is not the acid-resistance mechanism.

Q19. Gastric MALT lymphoma is notable because it:

- A) Never responds to treatment
- B) May regress after antibiotic eradication of H. pylori
- C) Is congenital
- D) Arises from parietal cells

Answer: B · Difficulty: Hard · LO 12.6

Why B: A lymphoma driven by chronic antigenic stimulation from the organism.

Why not: A, C, D are incorrect.

Q20. NSAIDs cause gastric injury because they:

- A) Are directly caustic only
- B) Stimulate gastrin
- C) Inhibit cyclo-oxygenase, removing prostaglandin support of mucus, bicarbonate and mucosal blood flow
- D) Kill H. pylori

Answer: C · Difficulty: Medium · LO 12.6

Why C: PGE2 maintains the mucosal defence.

Why not: A) A minor contribution. B, D) Incorrect.

Q21. Pyloric glands differ from fundic glands by having:

- A) Long glands with abundant parietal cells
- B) Goblet cells

C) No pits

D) Deep pits with short coiled mucous glands and few parietal cells

Answer: D · Difficulty: Medium · LO 12.3

Why D: Pit-to-gland ratio is the recognition key.

Why not: A) Fundus/body. B, C) Incorrect.

Q22. G cells secreting gastrin are located predominantly in the:

A) Antrum B) Fundus C) Duodenal submucosa D) Oesophagus

Answer: A · Difficulty: Medium · LO 12.3

Why A: Gastrinoma (Zollinger-Ellison) causes refractory ulceration.

Why not: B) Parietal and chief cells. C, D) Incorrect.

Q23. Chief cells secrete pepsinogen, which is activated by:

A) Trypsin B) Acid pH C) Bile D) Enterokinase

Answer: B · Difficulty: Medium · LO 12.3

Why B: Autocatalytic conversion to pepsin occurs at low pH.

Why not: A, D) Activate pancreatic zymogens. C) Emulsifies fat.

Q24. The appendix is recognised histologically by:

A) Villi B) Brunner glands C) A small, often irregular lumen with abundant lymphoid tissue encircling the wall D) Parietal cells

Answer: C · Difficulty: Easy · LO 12.5

Why C: Colonic-type mucosa with prominent circumferential lymphoid aggregates.

Why not: A, B, D belong to other regions.

Q25. The gastric mucosal barrier depends on all EXCEPT:

A) An adherent mucus-bicarbonate layer B) Tight junctions between surface cells C) Rapid epithelial turnover D) Absence of blood flow

Answer: D · Difficulty: Easy · LO 12.6

Why D: Rich mucosal blood flow is essential to remove back-diffused H⁺.

Why not: A, B, C are all genuine components.

Chapter 13 — Organs Associated with the Digestive System

Q1. In the classic hepatic lobule, blood flows:

- A) From the portal triads to the central vein, while bile flows the opposite way
- B) From the central vein to the portal triads
- C) In the same direction as bile
- D) Only through lymphatics

Answer: A · Difficulty: Easy · LO 13.1

Why A: Blood and bile flow in opposite directions — the key orientation fact.

Why not: B) Reversed. C) They are counter-current. D) Incorrect.

Q2. The portal triad contains:

- A) Central vein, artery, lymphatic
- B) Portal venule, hepatic arteriole, bile ductule
- C) Two veins and a nerve
- D) Sinusoid and canaliculus

Answer: B · Difficulty: Easy · LO 13.1

Why B: Plus lymphatics and nerves — the decisive feature identifying liver.

Why not: A, C, D are incorrect.

Q3. Centrilobular (zone 3) necrosis following shock occurs because zone 3 is:

- A) Nearest the portal triad B) Richest in glycogen C) Farthest along the oxygen gradient D) Avascular

Answer: C · Difficulty: Medium · LO 13.2

Why C: Lowest oxygen tension makes it most vulnerable to hypoxia — the "nutmeg liver."

Why not: A) That is zone 1. B, D) Incorrect.

Q4. Paracetamol toxicity also damages zone 3, but because that zone:

- A) Is hypoxic B) Is nearest the portal vein C) Lacks glutathione normally D) Has the highest cytochrome P450 activity, generating NAPQI

Answer: D · Difficulty: Hard · LO 13.2

Why D: Two mechanisms, same zone, different logic — a favourite examination discrimination.

Why not: A) The mechanism for ischaemic injury. C) Glutathione is depleted, not absent at baseline. B) Incorrect.

Q5. Hepatic stellate (Ito) cells normally store:

- A) Vitamin A B) Iron C) Glycogen D) Copper

Answer: A · Difficulty: Medium · LO 13.3

Why A: They hold about 80% of the body's vitamin B in lipid droplets, located in the space of Disse.

Why not: B) Kupffer cells/hepatocytes. C) Hepatocytes. D) Not a storage cell for copper.

Q6. The central cell of hepatic fibrosis is the:

A) Kupffer cell B) Activated stellate cell transdifferentiating into a myofibroblast C) Hepatocyte D) Pit cell

Answer: B · Difficulty: Hard · LO 13.3

Why B: It loses lipid droplets and secretes collagen into the space of Disse.

Why not: A) Macrophage. C) Parenchymal. D) NK cell.

Q7. Kupffer cells are located:

A) In the space of Disse B) In portal tracts only C) Within the sinusoidal lumen D) Inside hepatocytes

Answer: C · Difficulty: Medium · LO 13.3

Why C: Their intraluminal position distinguishes them from stellate cells and lets them clear portal-borne bacteria.

Why not: A) Stellate cells. B, D) Incorrect.

Q8. The space of Disse is significant because:

A) It contains bile

B) It stores glycogen

C) It is a lymphatic vessel

D) Plasma contacts hepatocyte microvilli directly across a discontinuous endothelium with deficient basal lamina

Answer: D · Difficulty: Hard · LO 13.3

Why D: This unrestricted access underlies the liver's synthetic and detoxifying role; hepatic lymph forms here.

Why not: A) Canaliculi. C) Lymph forms here but it is not a vessel. B) Intracellular.

Q9. The bile canaliculus is:

A) A groove between apposed apical membranes of two hepatocytes, sealed by tight junctions

B) A duct lined by cholangiocytes

C) A sinusoid

D) A lymphatic

Answer: A · Difficulty: Medium · LO 13.4

Why A: The hepatocyte is polarised, facing blood on one side and bile on the other.

Why not: B) Ductules, further downstream. C, D) Incorrect.

Q10. Obstructive (post-hepatic) jaundice produces:

- A) Unconjugated hyperbilirubinaemia
- B) Conjugated hyperbilirubinaemia with pale stools and dark urine
- C) Normal bilirubin
- D) Only raised transaminases

Answer: B · Difficulty: Medium · LO 13.4

Why B: Water-soluble conjugated bilirubin is excreted renally, darkening urine.

Why not: A) Haemolysis or conjugation defects. C, D) Incorrect.

Q11. Gilbert syndrome affects which step of bilirubin handling?

- A) Uptake B) Canalicular excretion C) Conjugation (reduced UGT1A1 activity) D) Enterohepatic recycling

Answer: C · Difficulty: Medium · LO 13.4

Why C: Mild unconjugated hyperbilirubinaemia, often manifest during fasting or illness.

Why not: A) Minor. B) Dubin-Johnson. D) Not the defect.

Q12. Dubin-Johnson syndrome produces a grossly black liver because of:

- A) Iron B) Copper C) Melanin D) Defective MRP2 canalicular excretion with pigment accumulation

Answer: D · Difficulty: Hard · LO 13.4

Why D: Conjugated hyperbilirubinaemia from an excretion defect.

Why not: A) Haemochromatosis. C) Not hepatic. B) Wilson disease.

Q13. Cirrhosis causes ascites and varices chiefly through:

- A) Architectural distortion obstructing portal flow, raising portal pressure
- B) Reduced albumin only
- C) Kupffer cell loss
- D) Bile duct proliferation

Answer: A · Difficulty: Medium · LO 13.5

Why A: The mechanical consequence; the synthetic consequence explains coagulopathy and hypoalbuminaemia.

Why not: B) Contributory but not primary for varices. C, D) Not the mechanism.

Q14. Centroacinar cells are unique to the:

- A) Parotid B) Pancreas C) Liver D) Gallbladder

Answer: B · Difficulty: Medium · LO 13.6

Why B: Intercalated duct cells extending into the acinar lumen — a decisive identifying feature.

Why not: A, C, D do not possess them.

Q15. Pancreatic bicarbonate secretion is performed by:

- A) Acinar cells B) Islet cells C) Duct and centroacinar cells via CFTR D) Stellate cells

Answer: C · Difficulty: Medium · LO 13.6

Why C: Acinar cells make enzymes; ducts make the alkaline fluid those enzymes require.

Why not: A) Enzymes. B) Hormones. D) Fibrosis.

Q16. In cystic fibrosis, exocrine pancreatic insufficiency results from:

- A) Autoimmune destruction
- B) Islet destruction
- C) Absent acinar cells from birth
- D) Viscid secretions obstructing ducts, causing autodigestion, atrophy and fibrosis

Answer: D · Difficulty: Medium · LO 13.6

Why D: Islets are relatively spared until late.

Why not: A) Type 1 diabetes mechanism. B, C) Incorrect.

Q17. Acute pancreatitis results fundamentally from:

- A) Premature intrapancreatic activation of trypsinogen initiating the enzyme cascade
- B) Bacterial infection
- C) Insulin deficiency
- D) Bile reflux only

Answer: A · Difficulty: Medium · LO 13.7

Why A: The zymogen safety system (enterokinase in duodenum; SPINK1 inhibitor) is bypassed.

Why not: B) Usually secondary. C) Unrelated. D) One possible trigger, not the mechanism.

Q18. Hypocalcaemia in acute pancreatitis results from:

- A) Parathyroid failure
- B) Calcium chelation by fatty acids released by lipase (saponification)
- C) Vitamin D deficiency
- D) Renal loss

Answer: B · Difficulty: Hard · LO 13.7

Why B: Fat necrosis produces chalky calcium soaps — a clinical sign traceable to histology.

Why not: A, C, D are not the mechanism.

Q19. The pancreas is distinguished from the parotid gland by the presence of:

- A) Striated ducts
- B) Adipose tissue
- C) Islets of Langerhans and centroacinar cells
- D) Serous acini

Answer: C · Difficulty: Medium · LO 13.8

Why C: The parotid has striated ducts and no islets; the pancreas has islets and no striated ducts.

Why not: A) Salivary. B) Salivary lobules. D) Both are serous — hence the confusion.

Q20. Striated ducts appear striated because of:

A) Collagen fibres B) Myofibrils C) Cilia D) Basal membrane infoldings with interposed mitochondria

Answer: D · Difficulty: Hard · **LO 13.9**

Why D: The arrangement supports active ion transport, modifying saliva.

Why not: A, B, C are incorrect.

Q21. The gallbladder wall is unique among hollow digestive organs in lacking:

A) A muscularis mucosae and submucosa B) Epithelium C) Smooth muscle D) Serosa entirely

Answer: A · Difficulty: Medium · **LO 13.10**

Why A: The decisive recognition feature, often with Rokitansky-Aschoff sinuses.

Why not: B, C) Both present. D) Partially covered by serosa.

Q22. The parotid gland is predominantly:

A) Mucous B) Serous C) Mixed mucous-predominant D) Sebaceous

Answer: B · Difficulty: Easy · **LO 13.9**

Why B: Submandibular is mixed serous-predominant; sublingual is mucous-predominant.

Why not: A, C) Other glands. D) Skin.

Q23. Sialolithiasis most commonly affects the submandibular gland because:

A) It is the largest
B) It is serous
C) Its secretion is more viscous and its duct has an uphill course
D) It lacks myoepithelial cells

Answer: C · Difficulty: Medium · **LO 13.9**

Why C: Both factors promote stasis and stone formation.

Why not: A) The parotid is larger. B) It is mixed. D) They are present.

Q24. Myoepithelial cells in salivary glands function to:

A) Secrete enzymes B) Produce mucin C) Absorb water D) Contract and expel secretion from acini

Answer: D · Difficulty: Easy · **LO 13.9**

Why D: They lie between the acinar cells and the basal lamina.

Why not: A, B) Acinar cells. C) Ducts.

Q25. The hepatic acinus of Rappaport is preferred to the classic lobule when the purpose is to:

A) Predict the zonal distribution of injury B) Describe bile drainage C) Identify the organ D) Count hepatocytes

Answer: A · Difficulty: Hard · LO 13.2

Why A: It is the functional unit defined by proximity to the arriving blood supply.

Why not: B) Portal lobule. C) Classic lobule. D) Not its purpose.

Chapter 14 — Respiratory System

Q1. A bronchiole is defined by the absence of:

- A) Smooth muscle B) Cartilage and glands C) Epithelium D) Elastic fibres

Answer: B · Difficulty: Easy · LO 14.1

Why B: The bronchus-to-bronchiole transition is defined by this loss, not by diameter.

Why not: A) Prominent in bronchioles. C, D) Present.

Q2. Goblet cells disappear at the bronchiolar level because:

- A) They are replaced by cilia
- B) There is no mucus requirement anywhere distally
- C) Mucus in cartilage-free airways would cause plugging and collapse
- D) They migrate to alveoli

Answer: C · Difficulty: Hard · LO 14.1

Why C: Bronchioles are splinted only by elastic recoil of surrounding parenchyma.

Why not: A, B, D are incorrect.

Q3. Club cells (formerly Clara cells) perform all of the following EXCEPT:

- A) Secrete surfactant-associated proteins and a protease inhibitor
- B) Detoxify xenobiotics via cytochrome P450
- C) Act as bronchiolar progenitor cells
- D) Produce mucin as their primary product

Answer: D · Difficulty: Medium · LO 14.2

Why D: They are specifically non-mucous, dome-shaped, non-ciliated cells.

Why not: A, B, C are all genuine functions.

Q4. The first structure in which gas exchange occurs is the:

- A) Respiratory bronchiole B) Terminal bronchiole C) Alveolar duct D) Trachea

Answer: A · Difficulty: Medium · LO 14.1

Why A: It has occasional alveoli budding from its wall; the terminal bronchiole is the last purely conducting airway.

Why not: B) Conducting. C) Further distal. D) Conducting.

Q5. The three layers of the blood-air barrier are:

- A) Three cell layers
- B) Type I pneumocyte, fused basal laminae, capillary endothelium
- C) Type II pneumocyte, surfactant, endothelium
- D) Macrophage, basal lamina, endothelium

Answer: B · Difficulty: Medium · LO 14.3

Why B: Thickness may be as little as 0.2 µm, in accordance with Fick's law.

Why not: A, C, D are incorrect.

Q6. Type II pneumocytes constitute approximately ____ of alveolar cells but cover ____ of the surface:

A) 40%; 95% B) 5%; 60% C) 60%; 5% D) 95%; 40%

Answer: C · Difficulty: Hard · LO 14.3

Why C: They are more numerous but far less extensive — the classic trap.

Why not: A) Describes type I cells. B, D) Incorrect.

Q7. Lamellar bodies within type II pneumocytes contain:

A) Mucin B) Lysosomal enzymes only C) Glycogen D) Surfactant

Answer: D · Difficulty: Easy · LO 14.3

Why D: Chiefly dipalmitoylphosphatidylcholine with surfactant proteins A–D.

Why not: A, B, C are incorrect.

Q8. Surfactant stabilises small alveoli because its tension-lowering effect:

A) Is greater at smaller radius as molecules pack more densely

B) Is uniform at all radii

C) Increases surface tension

D) Only acts in large alveoli

Answer: A · Difficulty: Hard · LO 14.4

Why A: By Laplace's law ($P = 2T/r$), small alveoli would otherwise empty into large ones.

Why not: B, C, D misstate the mechanism.

Q9. Neonatal respiratory distress syndrome shows histologically:

A) Emphysema B) Collapsed alveoli with eosinophilic hyaline membranes

C) Granulomas D) Bronchiectasis

Answer: B · Difficulty: Medium · LO 14.4

Why B: Proteinaceous exudate lines the collapsed airspaces.

Why not: A, C, D are other pathologies.

Q10. Which cell is the progenitor for alveolar epithelial repair?

A) Type I pneumocyte B) Alveolar macrophage C) Type II pneumocyte D) Club cell

Answer: C · Difficulty: Medium · LO 14.3

Why C: Type II hyperplasia is the histological sign of prior alveolar injury.

Why not: A) Terminally differentiated. B) Immune. D) Bronchiolar progenitor.

Q11. "Heart failure cells" are:

A) Cardiac myocytes B) Type II pneumocytes C) Fibroblasts D) Haemosiderin-laden alveolar macrophages

Answer: D · Difficulty: Medium · LO 14.5

Why D: They have ingested extravasated erythrocytes; demonstrated with Prussian blue.

Why not: A, B, C are incorrect.

Q12. Centriacinar emphysema from smoking predominantly affects the:

- A) Upper lobes, around respiratory bronchioles
- B) Lower lobes
- C) Pleura
- D) Trachea

Answer: A · Difficulty: Medium · LO 14.6

Why A: Smoke is most concentrated there; α 1-antitrypsin deficiency instead gives lower-lobe panacinar disease.

Why not: B) The α 1-antitrypsin pattern. C, D) Not emphysema sites.

Q13. Smoking promotes emphysema partly by:

- A) Increasing α 1-antitrypsin
- B) Recruiting neutrophils and oxidatively inactivating α 1-antitrypsin
- C) Reducing elastase
- D) Thickening alveolar walls

Answer: B · Difficulty: Hard · LO 14.6

Why B: It shifts the protease-antiprotease balance in both directions at once.

Why not: A, C, D are the reverse.

Q14. Loss of alveolar elastin causes airflow obstruction because it:

- A) Narrows the trachea
- B) Increases surfactant
- C) Removes elastic recoil and the radial traction splinting bronchioles, which then collapse on expiration
- D) Thickens the barrier

Answer: C · Difficulty: Hard · LO 14.6

Why C: The physiology is read directly from the lost elastin.

Why not: A, B, D are incorrect.

Q15. Curschmann spirals and Charcot-Leyden crystals are features of:

- A) Emphysema
- B) Pneumonia
- C) Tuberculosis
- D) Asthma

Answer: D · Difficulty: Medium · LO 14.6

Why D: Mucus casts of small airways and crystallised galectin-10 from eosinophils.

Why not: A, B, C show other findings.

Q16. The trachea is identified by:

- A) A C-shaped cartilage with trachealis muscle bridging the posterior gap
- B) Complete cartilage rings
- C) Absent cartilage

D) Stratified squamous epithelium

Answer: A · Difficulty: Easy · LO 14.7

Why A: The open posterior gap accommodates the oesophagus.

Why not: B) Not complete. C) Bronchioles. D) Wrong epithelium.

Q17. Olfactory epithelium is distinguished from respiratory epithelium by:

A) Goblet cells

B) Greater thickness, absence of goblet cells, and large underlying nerve bundles with Bowman glands

C) Cilia

D) Stratification

Answer: B · Difficulty: Hard · LO 14.7

Why B: The combination of thickness plus large nerve fascicles beneath is decisive.

Why not: A) Respiratory. C) Both have cilia (non-motile in olfactory). D) Both pseudostratified.

Q18. Alveolar macrophages are also known as:

A) Kupffer cells B) Langerhans cells C) Dust cells D) Microglia

Answer: C · Difficulty: Easy · LO 14.5

Why C: They phagocytose inhaled particulates and surfactant.

Why not: A) Liver. B) Skin. D) CNS.

Q19. Alveolar pores of Kohn permit:

A) Blood flow B) Surfactant secretion C) Lymph drainage D) Collateral air movement between adjacent alveoli

Answer: D · Difficulty: Medium · LO 14.3

Why D: They allow equalisation of pressure and collateral ventilation — but also spread of infection.

Why not: A, B, C are incorrect.

Q20. Lung parenchyma is distinguished from adipose tissue on low power by:

A) Capillary-containing septa and the presence of bronchioles with accompanying arteries

B) Colour

C) Empty appearance

D) Cell size

Answer: A · Difficulty: Medium · LO 14.7

Why A: Adipocytes have distinct outlines with peripheral nuclei and no airways.

Why not: B, C) Both look pale and empty — hence the confusion. D) Unreliable.

Q21. Pulmonary arteries accompany bronchi and bronchioles; pulmonary veins run:

A) Alongside the airway B) In the interlobular septa, away from airways C) In the pleura only D) Within alveolar walls

Answer: B · Difficulty: Hard · LO 14.7

Why B: A useful orientation rule on section.

Why not: A) Arteries. C, D) Incorrect.

Q22. The acinus of the lung comprises everything distal to a:

A) Bronchus B) Respiratory bronchiole C) Terminal bronchiole D) Alveolar sac

Answer: C · Difficulty: Medium · LO 14.1

Why C: The functional respiratory unit.

Why not: A) Too proximal. B, D) Components within it.

Q23. Antenatal corticosteroids reduce neonatal RDS by:

A) Increasing lung size B) Dilating airways C) Reducing inflammation only D) Accelerating type II pneumocyte maturation and surfactant production

Answer: D · Difficulty: Medium · LO 14.4

Why D: Directly targets the deficient cell.

Why not: A, B, C are not the mechanism.

Q24. The epithelium of the trachea is:

A) Pseudostratified ciliated columnar with goblet cells B) Simple squamous C) Stratified squamous D) Simple cuboidal

Answer: A · Difficulty: Easy · LO 14.7

Why A: Adapted for mucociliary clearance, over a conspicuously thick basement membrane.

Why not: B) Alveoli. C) Oesophagus. D) Small bronchioles.

Q25. Moving distally along the airway, smooth muscle becomes ____ relative to wall thickness while cartilage ____:

A) Less prominent; increases B) More prominent; disappears C) Unchanged; unchanged D) Absent; increases

Answer: B · Difficulty: Medium · LO 14.1

Why B: This is why bronchioles are the site of bronchoconstriction in asthma.

Why not: A, C, D reverse or deny the gradient.

Q26. Which statement about pneumocytes is correct?

A) Type II cells are the more numerous (~60% of cells) but cover only ~5% of the surface

B) Type I cells are both the more numerous and cover more surface

C) Type II cells cover ~95% of the surface

D) Type I and type II are present in equal numbers and cover equal areas

Answer: A · Difficulty: Hard · LO 14.3

Why A: Type I cells are ~40% of alveolar cells yet spread over ~95% of the surface because each is paper-thin; type II cells are ~60% of cells but compact and cuboidal, covering ~5%.

Why not: B) Type I cover more surface but are fewer. C) That is the type I figure. D) Neither number nor area is equal. Quoting only one of the two figures is what makes this a reliable trap.

Chapter 15 — Skin

Q1. The correct order of epidermal layers from base to surface is:

- A) Corneum, lucidum, granulosum, spinosum, basale
- B) Basale, granulosum, spinosum, corneum
- C) Basale, spinosum, granulosum, lucidum, corneum
- D) Spinosum, basale, granulosum, corneum

Answer: C · Difficulty: Easy · LO 15.1

Why C: Cells are produced basally and differentiate upward over roughly four weeks.

Why not: A) Reversed. B, D) Misordered.

Q2. Keratohyalin granules contain filaggrin, which functions to:

- A) Cross-link lipid
- B) Produce melanin
- C) Form desmosomes
- D) Aggregate keratin filaments into dense bundles

Answer: D · Difficulty: Medium · LO 15.2

Why D: This collapses the cell into a flattened, mechanically tough corneocyte.

Why not: A) Lamellar body lipids. B, C) Other structures.

Q3. Lamellar bodies discharge their contents in the stratum granulosum to:

- A) Form the intercellular lipid "mortar" preventing water loss
- B) Produce keratin
- C) Create desmosomes
- D) Transfer melanin

Answer: A · Difficulty: Medium · LO 15.2

Why A: The barrier is a brick-and-mortar structure; the lipid provides waterproofing.

Why not: B, C, D are incorrect.

Q4. Filaggrin (FLG) loss-of-function mutations are the strongest known genetic risk factor for:

- A) Psoriasis B) Atopic dermatitis (and cause ichthyosis vulgaris) C) Melanoma D) Pemphigus

Answer: B · Difficulty: Hard · LO 15.3

Why B: A defective barrier allows allergen entry and sensitisation — the "outside-in" hypothesis.

Why not: A, C, D have different mechanisms.

Q5. Parakeratosis in psoriasis means:

- A) Absent keratin
- B) Thickened granular layer
- C) Retention of nuclei in the stratum corneum
- D) Increased melanin

Answer: C · Difficulty: Medium · LO 15.3

Why C: Accelerated transit (~4 days) means cells reach the surface before losing their nuclei.

Why not: A, B, D are incorrect; the granular layer is lost.

Q6. The Auspitz sign (pinpoint bleeding on scale removal) in psoriasis results from:

- A) Vasculitis B) Infection C) Platelet defect D) Dilated capillaries in elongated dermal papillae close to a thinned suprapapillary plate

Answer: D · Difficulty: Hard · LO 15.3

Why D: A clinical sign read directly from the histology.

Why not: A, B, C are incorrect.

Q7. Melanocytes are derived from:

- A) Neural crest B) Mesoderm C) Endoderm D) Bone marrow

Answer: A · Difficulty: Easy · LO 15.4

Why A: They migrate to the basal epidermis during development.

Why not: B, C) Wrong germ layer. D) Langerhans cells.

Q8. Racial variation in skin colour is due mainly to differences in:

- A) Melanocyte number
- B) Melanosome size, melanisation, dispersion and degradation rate
- C) Keratinocyte number
- D) Epidermal thickness

Answer: B · Difficulty: Hard · LO 15.4

Why B: Melanocyte density is essentially the same across populations.

Why not: A) A common misconception. C, D) Not the basis.

Q9. Albinism differs from vitiligo in that albinism features:

- A) Absent melanocytes
- B) Autoimmune destruction
- C) Normal melanocyte number with defective tyrosinase or melanosome biogenesis
- D) Patchy depigmentation only

Answer: C · Difficulty: Medium · LO 15.4

Why C: Number versus function is the discriminator; vitiligo shows absent melanocytes in affected patches.

Why not: A, B, D describe vitiligo.

Q10. Melanin protects DNA by accumulating in keratinocytes:

- A) Randomly in the cytoplasm
- B) In lysosomes only
- C) In the nucleus
- D) As a cap over the nucleus on the UV-facing side

Answer: D · Difficulty: Medium · LO 15.4

Why D: A UV shield positioned precisely where DNA needs protection.

Why not: A, B, C are incorrect.

Q11. Langerhans cells are:

- A) Bone-marrow-derived antigen-presenting dendritic cells with Birbeck granules
- B) Melanin-producing
- C) Mechanoreceptors
- D) Keratinocyte precursors

Answer: A · Difficulty: Medium · LO 15.4

Why A: Located mainly in the stratum spinosum; central to contact dermatitis.

Why not: B) Melanocytes. C) Merkel cells. D) Basal keratinocytes.

Q12. Merkel cells function as:

- A) Immune cells
- B) Mechanoreceptors for sustained light touch and texture
- C) Pigment cells
- D) Secretory cells

Answer: B · Difficulty: Medium · LO 15.4

Why B: They associate with an expanded nerve terminal in the basal layer.

Why not: A) Langerhans. C) Melanocytes. D) Glandular cells.

Q13. Thick skin is distinguished from thin skin by:

- A) A thicker dermis
- B) More melanocytes
- C) A stratum lucidum, very thick corneum, and absence of hair follicles and sebaceous glands
- D) Presence of apocrine glands

Answer: C · Difficulty: Medium · LO 15.5

Why C: "Thick/thin" refers to epidermal thickness — the dermis of the back is thicker than that of the palm.

Why not: A) A common trap. B, D) Incorrect.

Q14. The Pacinian corpuscle is recognised by:

- A) Stacked flattened cells in a dermal papilla
- B) A spindle shape
- C) Free nerve endings
- D) Large oval body of concentric onion-like lamellae in the deep dermis/hypodermis

Answer: D · Difficulty: Easy · LO 15.6

Why D: The lamellae filter out sustained pressure, making it a vibration detector.

Why not: A) Meissner corpuscle. C) Nociceptors. B) Ruffini.

Q15. Meissner corpuscles are located in the:

A) Dermal papillae B) Hypodermis C) Stratum corneum D) Muscle

Answer: A · Difficulty: Medium · LO 15.6

Why A: Positioned superficially for light touch and low-frequency vibration.

Why not: B) Pacinian. C) Dead layer. D) Muscle spindles.

Q16. Eccrine sweat gland ducts are identified by their:

A) Single layer of columnar cells
B) Distinctive two-layered cuboidal epithelium
C) Stratified squamous lining
D) Pseudostratified lining

Answer: B · Difficulty: Medium · LO 15.7

Why B: The double-layered duct is the decisive recognition feature.

Why not: A, C, D are incorrect.

Q17. The sweat test in cystic fibrosis is abnormal because of failure of:

A) Sweat production
B) Myoepithelial contraction
C) Ductal chloride reabsorption
D) Sodium secretion in the coil

Answer: C · Difficulty: Hard · LO 15.7

Why C: Defective CFTR prevents Cl⁻ reabsorption, so sweat chloride is high.

Why not: A, B, D are incorrect.

Q18. Apocrine glands differ from eccrine glands by:

A) Being distributed over the whole body
B) Lacking myoepithelial cells
C) Producing watery sweat for thermoregulation
D) Opening into hair follicles, being restricted to axilla/areola/anogenital areas, and developing at puberty

Answer: D · Difficulty: Medium · LO 15.7

Why D: Their secretion is odourless until bacterial action.

Why not: A, C) Eccrine. B) Present.

Q19. Sebaceous glands secrete by which mechanism?

A) Holocrine B) Apocrine C) Merocrine D) Endocrine

Answer: A · Difficulty: Easy · LO 15.7

Why A: The entire cell disintegrates to become sebum.

Why not: C) Exocytosis. B) Apical pinching. D) No duct.

Q20. The papillary dermis differs from the reticular dermis by having:

A) Thick collagen bundles

B) Loose connective tissue with fine collagen, capillary loops and Meissner corpuscles

C) No vessels

D) Only elastic fibres

Answer: B · Difficulty: Medium · LO 15.8

Why B: Interdigitation with rete ridges increases adhesion and resists shear.

Why not: A) Reticular dermis. C, D) Incorrect.

Q21. The epidermis is nourished by:

A) Its own capillaries

B) Lymphatics within the epidermis

C) Diffusion from capillaries in the papillary dermis

D) Sweat

Answer: C · Difficulty: Easy · LO 15.8

Why C: Epidermis, like all epithelium, is avascular.

Why not: A, B) No vessels or lymphatics within it. D) Not nutritive.

Q22. Skin is distinguished from oesophageal mucosa (both stratified squamous) by:

A) Cell shape

B) Basement membrane

C) Nuclear size

D) Keratinised anucleate surface plus dermal appendages

Answer: D · Difficulty: Medium · LO 15.9

Why D: The oesophagus is non-keratinised with nuclei to the surface, has no appendages, and has a muscularis mucosae.

Why not: A, B, C are not discriminating.

Q23. Melanocytes appear as clear cells in the basal layer on H&E because of:

A) A shrinkage artifact producing a perinuclear halo B) Lipid content C) Glycogen D) Mucin

Answer: A · Difficulty: Hard · LO 15.4

Why A: B useful artifact — it makes them findable.

Why not: B, C, D are incorrect.

Q24. Acne results from:

A) Eccrine gland blockage

- B) Sebaceous follicle obstruction with *Cutibacterium acnes* colonisation under androgenic stimulation
- C) Apocrine infection
- D) Melanocyte dysfunction

Answer: B · Difficulty: Medium · LO 15.7

Why B: Explains its onset at puberty and follicular distribution.

Why not: A, C, D are incorrect.

Q25. Desmosomes are most prominent in which epidermal layer, giving it its name?

- A) Basale B) Granulosum C) Spinosum D) Corneum

Answer: C · Difficulty: Easy · LO 15.1

Why C: The "prickles" are desmosomal contacts persisting where processing shrinkage separates cells.

Why not: A) Hemidesmosomes predominate. B, D) Later differentiation stages.

Chapter 16 — Endocrine System

Q1. Tubular mitochondrial cristae and abundant smooth ER indicate a cell secreting:

A) Peptide hormone B) Mucus C) Catecholamine D) Steroid hormone

Answer: D · Difficulty: Medium · LO 16.1

Why D: Steroidogenic P450 enzymes reside in SER and tubular-cristae mitochondria.

Why not: A) RER, Golgi, granules. C) Dense-core granules. B) Exocrine.

Q2. Steroid hormones are not stored in secretory granules because they:

A) Are lipid-soluble and diffuse through membranes B) Are too large C) Are toxic D) Degrade rapidly

Answer: A · Difficulty: Hard · LO 16.1

Why A: They must therefore be synthesised on demand, making steroid responses slower than peptide responses.

Why not: B, C, D are incorrect.

Q3. The anterior pituitary develops from:

A) Neuroectoderm B) Oral ectoderm (Rathke's pouch) C) Mesoderm D) Endoderm

Answer: B · Difficulty: Medium · LO 16.2

Why B: A true glandular epithelium, unlike the neural posterior lobe.

Why not: A) Posterior pituitary. C, D) Incorrect.

Q4. The posterior pituitary:

A) Synthesises ADH and oxytocin
B) Contains acidophils and basophils
C) Stores and releases hormones made in the hypothalamus
D) Is controlled by a portal system

Answer: C · Difficulty: Easy · LO 16.2

Why C: Made in supraoptic and paraventricular nuclei and transported down axons.

Why not: A) It synthesises none. B) Anterior lobe. D) Neural, not vascular, control.

Q5. Herring bodies are:

A) Colloid droplets B) Secretory granules of acidophils C) Glial nuclei D) Dilated axon terminals with neurosecretory granules

Answer: D · Difficulty: Medium · LO 16.2

Why D: A decisive recognition feature of the neurohypophysis, along with pituicytes.

Why not: A) Thyroid. C) Pituicytes are the glia themselves. B) Anterior lobe.

Q6. Basophils of the anterior pituitary produce (mnemonic B-FLAT):

A) FSH, LH, ACTH, TSH B) GH and prolactin C) ADH and oxytocin D) Calcitonin

Answer: A · Difficulty: Medium · LO 16.2

Why A: Acidophils make GH and prolactin.

Why not: B) Acidophils. C) Posterior lobe. D) Thyroid C cells.

Q7. The hypophyseal portal system is so called because it has:

A) One capillary bed B) Two capillary beds in series C) No veins D) Arterial anastomoses only

Answer: B · Difficulty: Hard · LO 16.2

Why B: It delivers hypothalamic hormones at high concentration without systemic dilution.

Why not: A, C, D are incorrect.

Q8. Prolactin is unique among anterior pituitary hormones in being under predominantly:

A) Stimulatory control B) No control C) Inhibitory control by dopamine D) Direct neural control

Answer: C · Difficulty: Hard · LO 16.2

Why C: Stalk section therefore raises prolactin while lowering all other anterior pituitary hormones.

Why not: A, B, D are incorrect.

Q9. The thyroid is unique among endocrine glands in that it:

A) Has no blood supply
B) Lacks a capsule
C) Secretes only steroids
D) Stores its hormone extracellularly as colloid

Answer: D · Difficulty: Medium · LO 16.3

Why D: Thyroglobulin in the follicular lumen holds months of supply.

Why not: A, B, C are incorrect.

Q10. Thyroid peroxidase catalyses:

A) Oxidation of iodide and organification of tyrosine residues B) Iodide trapping C) Colloid endocytosis D) T4-to-T3 conversion in the periphery

Answer: A · Difficulty: Medium · LO 16.3

Why A: It is the target of thionamide antithyroid drugs and the antigen in Hashimoto thyroiditis.

Why not: B) NIS. C) Separate step. D) Deiodinases.

Q11. The sodium-iodide symporter (NIS) is inhibited by:

A) Carbimazole B) Perchlorate C) Propranolol D) Levothyroxine

Answer: B · Difficulty: Hard · LO 16.3

Why B: It competes for the transporter; radioiodine uptake also exploits NIS.

Why not: A) Inhibits TPO. C) Symptomatic. D) Replacement.

Q12. Scalloped resorption vacuoles at the colloid margin with tall columnar follicular cells indicate:

A) An inactive gland B) Neoplasia C) An actively stimulated gland D) Fibrosis

Answer: C · Difficulty: Hard · LO 16.3

Why C: A direct visual diagnosis of hyperstimulation, as in Graves disease.

Why not: A) Flat cells, dense distended colloid. B, D) Different features.

Q13. Current lineage-tracing evidence indicates thyroid C cells derive from:

A) Neural crest B) Surface ectoderm C) Mesoderm D) Pharyngeal endoderm via the ultimobranchial body

Answer: D · Difficulty: Hard · LO 16.3

Why D: Sox17-Cre lineage tracing shows an endodermal origin, overturning the older neural crest teaching (which many examinations still use).

Why not: A) The classical but superseded answer. B, C) Incorrect.

Q14. Medullary thyroid carcinoma arises from C cells and characteristically shows:

A) Amyloid stroma derived from calcitonin B) Psammoma bodies C) Colloid-filled follicles D) Hürthle cells

Answer: A · Difficulty: Medium · LO 16.3

Why A: It occurs in MEN 2B and 2A and secretes calcitonin.

Why not: B) Papillary carcinoma. C) Normal/follicular. D) Hashimoto/Hürthle cell tumour.

Q15. PTH stimulates osteoclasts:

A) Directly via osteoclast PTH receptors
B) Indirectly, by acting on osteoblasts to increase RANKL
C) Through calcitonin
D) Via vitamin D only

Answer: B · Difficulty: Hard · LO 16.4

Why B: Osteoclasts lack PTH receptors — a classic trap.

Why not: A) False. C) Opposes PTH. D) One of several PTH effects.

Q16. Primary hyperparathyroidism produces:

A) High calcium, high phosphate B) Low calcium, low phosphate C) High calcium, low phosphate D) Low calcium, high phosphate

Answer: C · Difficulty: Medium · LO 16.4

Why C: PTH raises calcium while promoting renal phosphate excretion.

Why not: A, B, D are inconsistent with PTH action.

Q17. Oxyphil cells of the parathyroid are:

- A) The source of PTH
- B) Chief cell precursors only
- C) Adipocytes
- D) Large eosinophilic mitochondria-rich cells of uncertain function, increasing after puberty

Answer: D · Difficulty: Medium · LO 16.4

Why D: Chief cells produce PTH.

Why not: A) Chief cells. B, C) Incorrect.

Q18. The zona fasciculata secretes ____ and appears ____:

- A) Cortisol; pale and vacuolated (spongiocytes)
- B) Aldosterone; dark
- C) Androgens; eosinophilic
- D) Catecholamines; basophilic

Answer: A · Difficulty: Easy · LO 16.5

Why A: It holds the most lipid (cholesterol ester precursor), extracted in processing.

Why not: B) Glomerulosa. C) Reticularis. D) Medulla.

Q19. Aldosterone secretion by the zona glomerulosa is regulated chiefly by:

- A) ACTH
- B) Renin-angiotensin and potassium
- C) TSH
- D) Dopamine

Answer: B · Difficulty: Medium · LO 16.5

Why B: This is why glomerulosa function is relatively preserved in isolated ACTH deficiency.

Why not: A) Fasciculata and reticularis. C, D) Incorrect.

Q20. The adrenal medulla produces mostly adrenaline (rather than noradrenaline) because:

- A) It lacks noradrenaline
- B) It is innervated postganglionically
- C) Cortisol arriving from the cortex induces PNMT
- D) It is neural crest derived

Answer: C · Difficulty: Hard · LO 16.5

Why C: The corticomedullary blood flow gradient explains why extra-adrenal chromaffin tissue makes noradrenaline instead.

Why not: A) It makes both. B) Innervated preganglionically. D) True but not the reason.

Q21. Adrenal medullary chromaffin cells are innervated by:

- A) Postganglionic sympathetic fibres
- B) Somatic motor fibres
- C) Parasympathetic fibres
- D) Preganglionic sympathetic fibres

Answer: D · Difficulty: Medium · LO 16.5

Why D: They are modified postganglionic neurons that have lost their axons, so they respond instantly.

Why not: A, B, C are incorrect.

Q22. In 21-hydroxylase deficiency, adrenal hyperplasia occurs because:

- A) Cortisol deficiency removes negative feedback, raising ACTH
- B) ACTH is suppressed
- C) Aldosterone is excessive
- D) The adrenal is infiltrated

Answer: A · Difficulty: Hard · LO 16.5

Why A: Precursors are shunted to androgens, causing virilisation, with salt wasting if aldosterone is also deficient.

Why not: B) ACTH rises. C) It is deficient. D) Not the mechanism.

Q23. Beta cells constitute about ____ of islet cells and are located ____:

- A) 20%; peripherally B) 70%; centrally C) 5%; scattered D) 50%; peripherally

Answer: B · Difficulty: Medium · LO 16.6

Why B: Alpha cells (~20%) are peripheral; delta cells are scattered.

Why not: A) Alpha. C) Delta. D) Incorrect.

Q24. Islet amyloid in type 2 diabetes is derived from:

- A) Insulin B) Calcitonin C) Islet amyloid polypeptide (amylin), co-secreted with insulin D) Immunoglobulin light chains

Answer: C · Difficulty: Hard · LO 16.6

Why C: Type 1 diabetes instead shows insulinitis with beta-cell depletion.

Why not: A) Not directly. B) Medullary thyroid carcinoma. D) AL amyloidosis.

Q25. Adrenal cortex may be confused with liver on low power; the decisive discriminator is:

- A) Cell colour
- B) Nuclear shape
- C) Cell size
- D) A thick capsule with zonation and lipid-rich vacuolated cells versus portal triads and binucleate hepatocytes

Answer: D · Difficulty: Medium · LO 16.7

Why D: Find a portal triad (liver) or a zoned cortex beneath a capsule (adrenal).

Why not: A) Both form eosinophilic cords — hence the confusion. B, C) Unreliable.

End of Question Bank

Total: 417 questions across the 16 official chapters.

Every item is traceable to chapter → learning objective → answer → rationale → difficulty, and every stated learning objective is assessed. Items were reviewed to remove over-specific molecular-pathology claims, to ensure each distractor is a plausible and instructive error, and to balance the answer key across all four options.